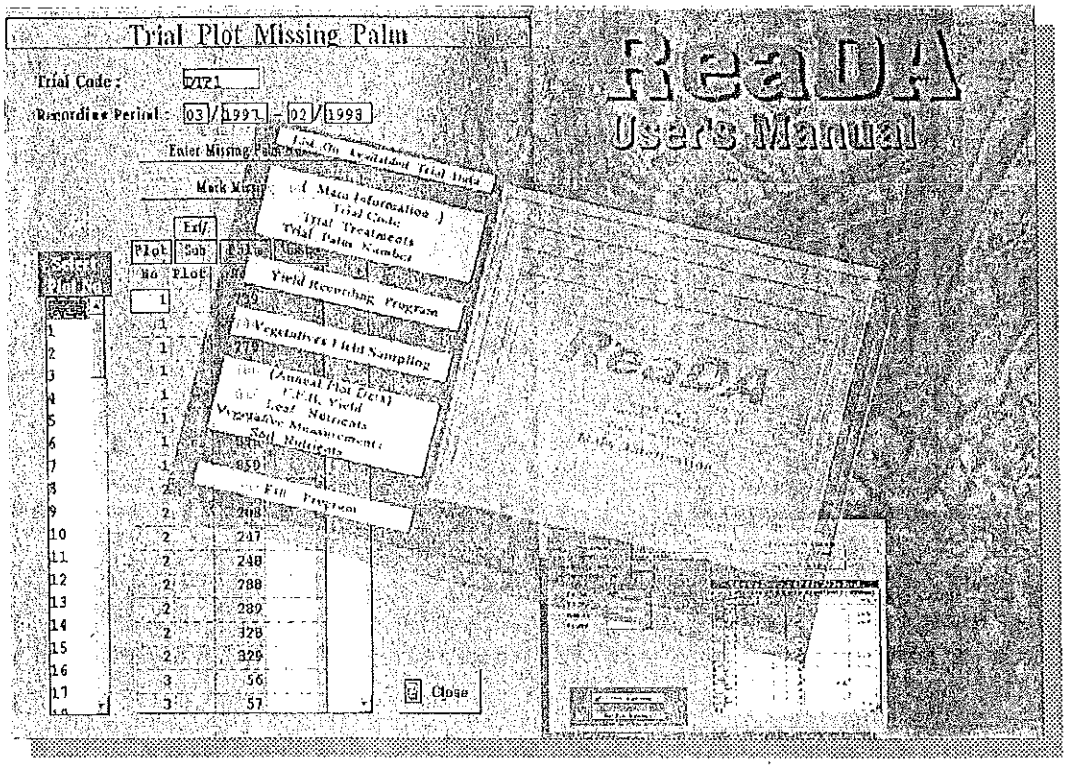
ReaDa

Relational Database For Agronomy Trial Data



PALM OIL RESEARCH INSTITUTE OF MALAYSIA

PORIM

MPOB, LAHAD DATU

AHMAD AFANDI B. MURDI
Pegawai Penyelidik
Stesen MPOB Lahad Datu.

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Requirements for ReaDA

Following are recommended requirements for running ReaDA with Windows 95/98:

- An IBM - compatible computer with a Pentium or equivalent 133 MHz processor (or higher).
- 4 MB of hard disk free space for installation more free space is required for data file.

Installation

ReaDA contains three installation diskettes: ReaDA1, ReaDA2, ReaDA3. To install ReaDA:

1. Insert diskette ReaDA1 in drive A:
2. From start menu, choose *Run.*
3. Type a:\setup and press *Enter*

ReaDA will be installed in directory C:\Reada

Default ReaDA Data File

The following are data file that will be created in Reada directory and used by the ReaDA system:

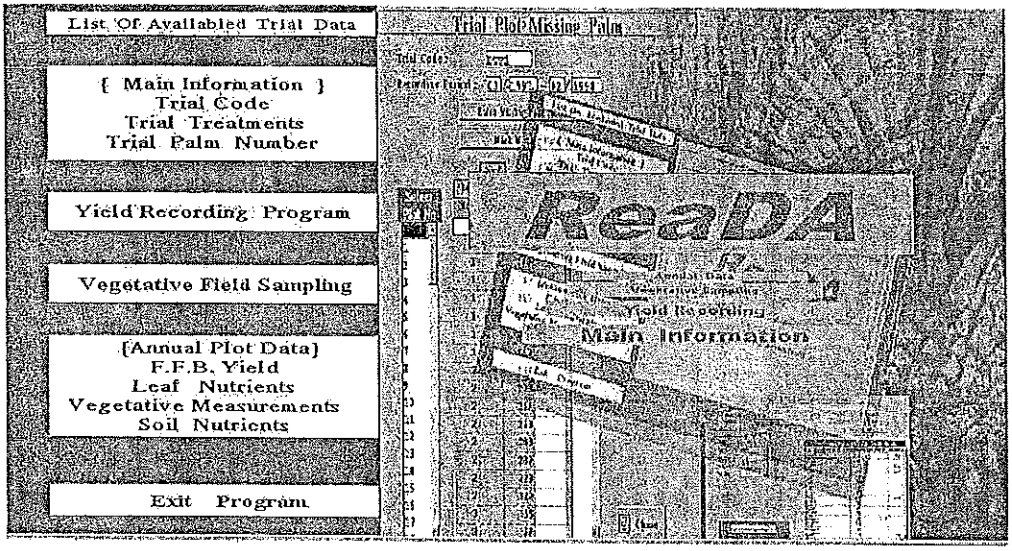
- acfsdata - contain data file structure
- datainfo - contain trial information
- datatmnt - contain trial treatment
- datapalm - contain trial palm number
- dataryld - contain trial recording yield
- dataffb - contain trial annual yield data
- dataleaf - contain trial leaf nutrients data
- datavm - contain trial vegetative measurement data
- datasoil - contain trial soil nutrients data
- dataveg - contain trial vegetative sampling data
- acfsinfo - contain trial info file structure

* all file that is being created in ReaDA system by default in Reada directory should not be modified.

INTRODUCTION

This manual, the user's guide, explains things you need to know in using ReaDA (Relational Database for Agronomy Trial Data). ReaDA is developed using Borland's Visual Dbase running in Windows 95/98. No matter what level of experience you have with database products, it will help you enter the oil palm yield recording data at ease.

Figure 1: Relational Database for Agronomy Trial Data (ReaDA)



Welcome to READA

To start READA

1) If you have not started your Windows, do so now

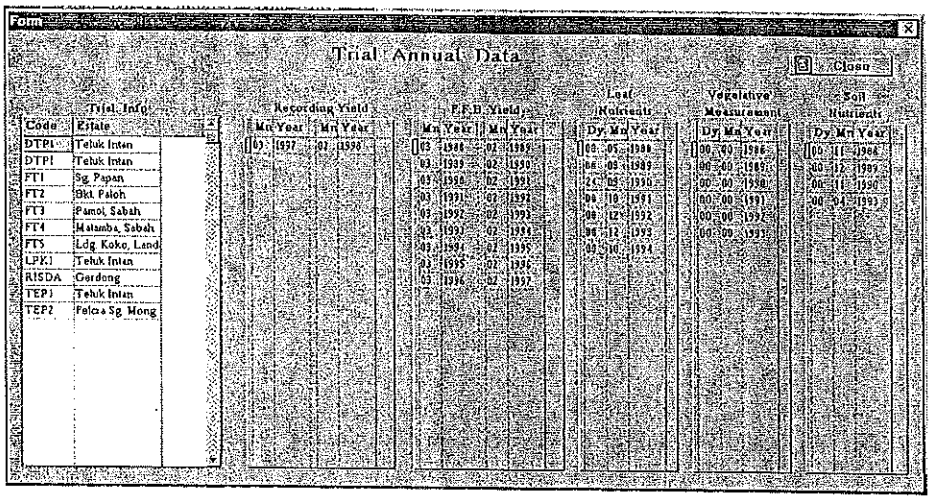
2) Double-click the **READA** icon.

- Your computer screen should look like the Figure 1. The menu bar consists of several operations for the data entry.

Lesson1

LIST OF AVAILABLE TRIAL DATA

Figure 2: Trial Annual Data



If you want to check the data that are already saved in the program, you can go through it easily.

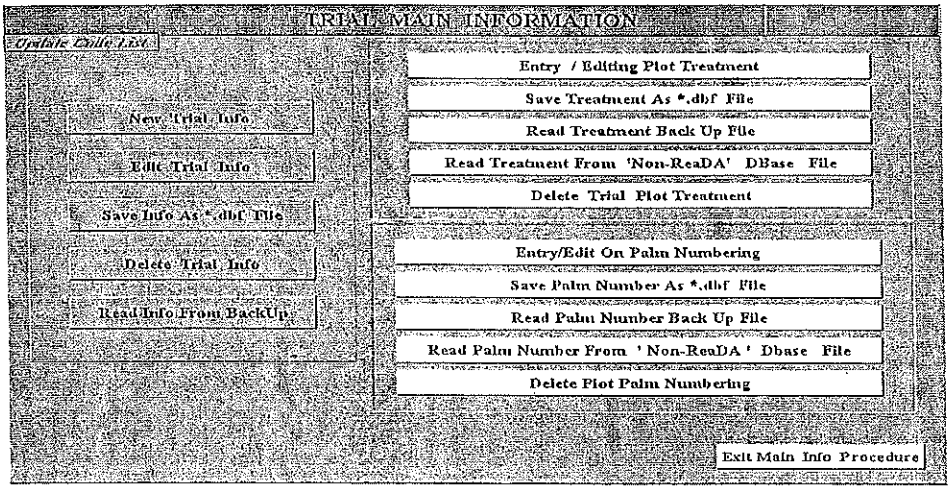
The steps are as follows:

1) Click to *'List Of Available Trial Data'* icon and the *'Trial Annual Data'* will appear.
   - Your screen would look like Figure 2.
   - Here, you can see the listed trial information, the duration of recording yield and FFB yield, and also the sampling date of leaf nutrients, vegetative measurement and soil nutrients.
   - This list will automatically exist when you have created and saved any trial data.

Lesson 2

{MAIN INFORMATION} TRIAL CODE, TRIAL TREATMENTS, TRIAL PALM NUMBER

Figure 3: Trial Main Information



This menu is used when you want to create a new main information for trial code, trial treatment and trial palm number.

The step are as follows:

1) Click *'{Main Information} Trial Code, Trial Treatments, Trial Palm Number'* icon.
   - Your screen would look like Figure 3

2.1 Creating a New Trial Information

This procedure is used when you want to create a new main information for the trial.

The following steps are to be taken:

1) Click *'New Trial Info'* button and the *'Trial Main Information'* will appear on the screen.

2) Key in your *Trial Code*:

e.g. DTP1

3) *Estate*: Enter your trial location or estate name

e.g. Teluk Intan

4) *Planting Year*: Enter the planting year of your trial

e.g. 1988

5) Enter your *Total No. Of Treatment Factor*, e.g. 6 and state the *Label* of any treatment under the label column.

e.g. Rep, D, N, P, K, L

6) Key in the detailed explanation of the treatment label in the *'Treatment Factor Information'* column if necessary.

e.g. Rep = Replicate, D = Density, N = Nitrogen,
K = Potassium, L = Lime

7) Click *'Yes'* button if you want to save the entered data or *'No'* if you want to ignore it.

2.2 Editing The Trial Info

This procedure is used when the trial information entries need to be changed.

The following steps are to be taken:

1) Click *'Edit Trial Info'* button and the *'Entry/Editing Trial Code'* will appear.

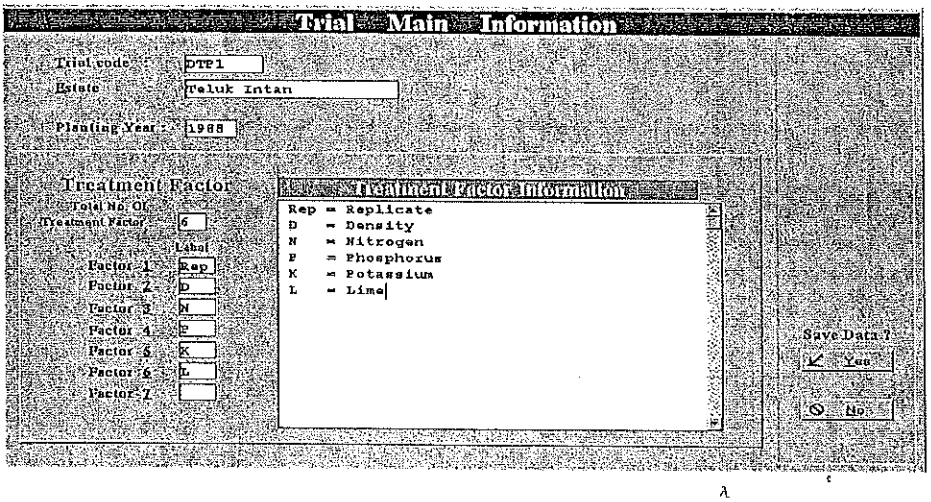
2) *Select Code For Editing* by clicking the mouse.

e.g. DTP1 Teluk Intan

3) Click *'Yes/Continue'* button to continue the editing procedure otherwise click *'Cancel-Exit'* button.

- If 'Yes/Continue', the *'Trial Main Information'* will appear and it would look like Figure 4.

Figure 4: Edit Trial Info (Trial Main Information)



4) Now, you can edit any information by typing directly into the column.

5) Save your editing by clicking *'Yes'* or otherwise click *'No'* button to quit the procedure without saving your entered data.

2.3 Saving Trial Info As a Database File

This procedure is used when you want to save the trial information that you have created before.

The following steps are to be taken:

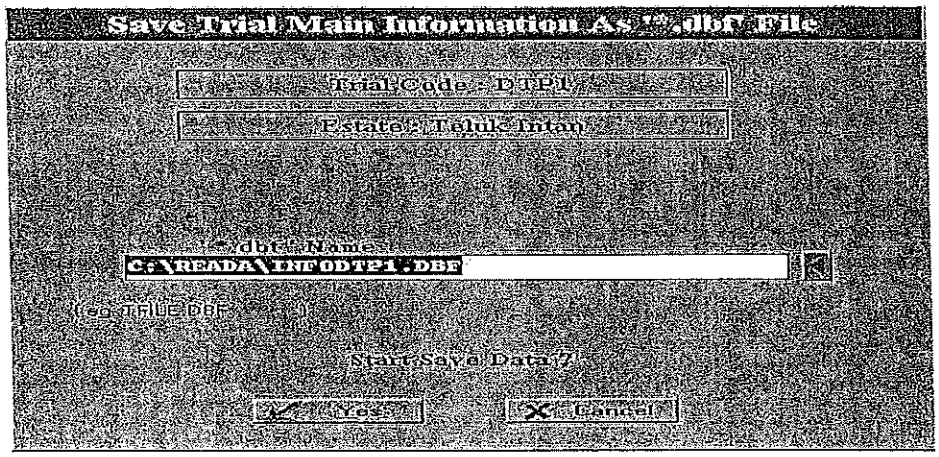
1) Click *'Save Info As *.dbf File'* button and the *'Save Trial Info As *.dbf File'* will appear.

2) Now, *Select Code To Be Saved* by clicking the mouse to the selection.

3) Click *'Yes/Continue'* button to continue the procedure or otherwise click *'Cancel-Exit'*.

- If you decide to continue, the *'Save Trial Main Information As *.dbf File'* will appear and look like Figure 5.

Figure 5: SaveTrial Main Information As *.dbf File



4) Enter the *'*.dbf ' Name:* with its drive and the directory or you can browse the listed files by clicking  toolbar.

e . g . C : \READA\INFODTP1.DBF

Note: File will be automatically saved into the .dbf file

The filename must be less than eight characters.

5) Click *'Yes'* if you want to confirm to save your file or otherwise click *'Cancel'*.

2.4 Deleting Trial Information

With this procedure, you may delete your data records that you have saved permanently.

Steps:

1) Click ***'Delete Trial Info'*** button and ***'Deleting Trial Code'*** will appear.

2) ***Select Code To Be Deleted*** by clicking the selection.

e.g. DTP1 Teluk Intan

3) Click *'Yes/Continue'* to delete trial information, otherwise click *'Cancel-Exit'*.

4) You can confirm your deleting process by clicking *'Yes'* button or click *'Cancel'* button to deny your request.

- If the main information has had their own data record, you have to ensure that every data for the trial is already deleted before deleting the main information can be done.

2.5 Reading Trial Information From Back Up File

This procedure is used when the users need to recover the trial data especially when they have deleted the particular data.

Steps:

1) Click *'Read Info From Back Up'* button and the *'Read Trial Main Info From Back Up File'* will appear.

2) Enter your ***Back Up File Name*** including the drive and the directory if necessary. e.g. C: \READA\BCINFO.DBF

- You can browse the directories and double-click to the selected file name.

3) Click *'View Information In Back Up File'* button.

- Now, you can see the information available in the back up file.

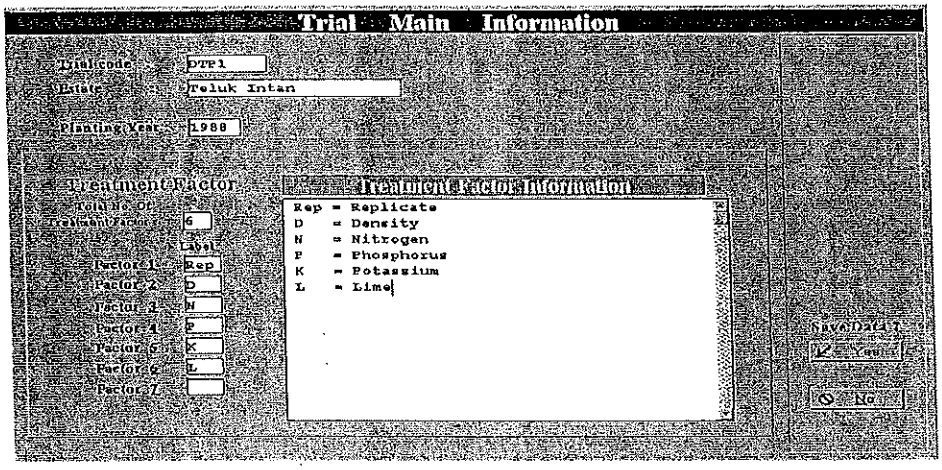
- The program cannot read the data when the selected file is not of the same type with the data to be read.

4) *Select Code To Be Read* from the list by clicking your mouse.

e.g. DTP1 Teluk Intan

5) Click *'Start Replacing Data From Back Up File'* button and your screen would look like Figure 6 or otherwise click *'Exit Read Back Up'*.

Figure 6: Trial Main Information (Read Info From Back Up)



6) Click either *'Yes'* button to save your data or *'No'* if not.

2.6 Entering or Editing Plot Treatment

You can enter a new data for plot treatment or edit the data in the same form and procedures.

Steps:

1) Click *'Entry/Editing Plot Treatment'* button and *'Editing Trial Treatment'* will appear.

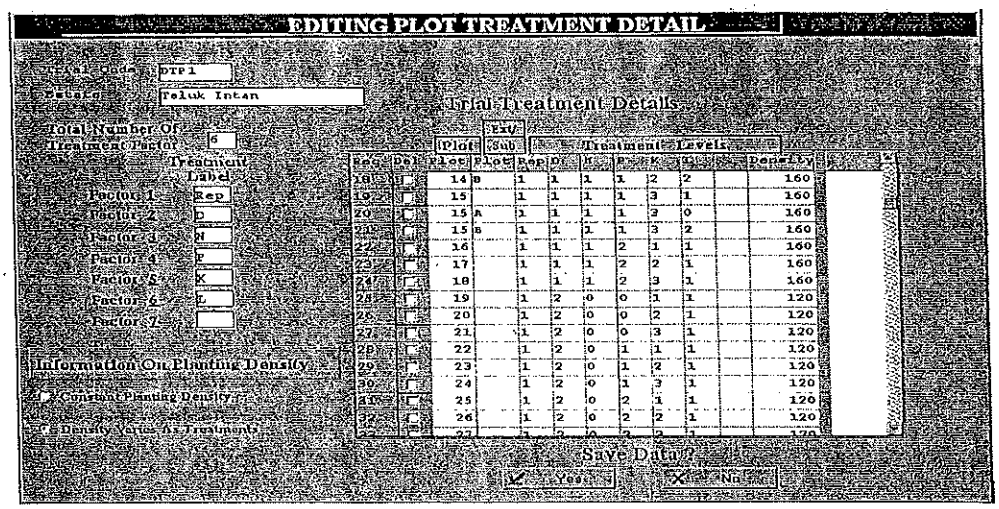
2) *Select Code For Editing* by clicking the mouse.

e.g. DTP1 Teluk Intan

3) Click *'Yes/Continue'* button to continue, otherwise click *'Cancel-Exit'*.

- If you click 'Yes/Continue', *'Editing Plot Treatment Detail'* form will appear as shown in Figure 7.

Figure 7: Editing Plot Treatment Detail



4) Choose your *Planting Density.*

- Mark the *'Constant Planting Density'* if the number of planting densities is constant in every plot area.
  State the number into the column beside.
- Mark the *'Densities Varies As Treatment'* if the number of planting densities varies in every plot area.
  You can enter the number under the *'Density'* column directly.

5) If you click any box under the *'Del'* column, the row will be deleted.

6) *Save* your entry/edited data.

2.7 Saving Plot Treatment As a Database File

This procedure is used when you need to save your plot treatment to a database file.

Steps:

1) Click *'Save Treatment As *.dbf File'* button and *'Save Treatment As *.dbf File'* form will appear.

2) *Select Code To Be Saved* by clicking your mouse.

e.g. DTP1 Teluk Intan

3) Confirm your selection by clicking *'Yes/Continue'* button to continue or *'Cancel-Exit'* to quit the saving procedure.

- If 'Yes/Continue', your screen will show *'Save Plot Treatment Data As *.dbf' File'* form as in Figure 8.

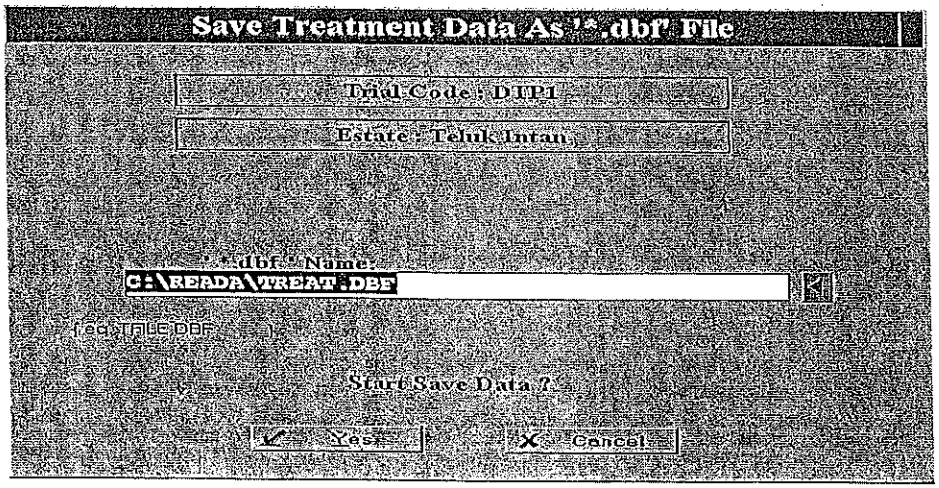
4) Enter the *'*.dbf' Name:* or select from the listed file.

e.g. C: \READA\TREAT.DBF

(Drive\Dir\Filename.dbf)

5) Save your file by clicking *'Yes'* button or *'Cancel'* button to quit without saving it.

Figure 8: Save Treatment Data As *.dbf' File



2.8 Reading Treatment From Backup File

This procedure is used when you have to recover the treatment data record especially when the file is lost.

Steps:

1) Click *'Read Treatment Backup File'* button and your screen will show *'Read Treatment From Backup File'*.

2) *Select Code To Be Read* by clicking your mouse

   e.g. DTP1 Teluk Intan

3) Confirm your selection by clicking *'Yes/Continue'* button.

   - Your screen will show *'Read Treatment Data From Backup File'*.

4) Enter the *Back Up File Name* with its drive and the directory if necessary or select by browse template.

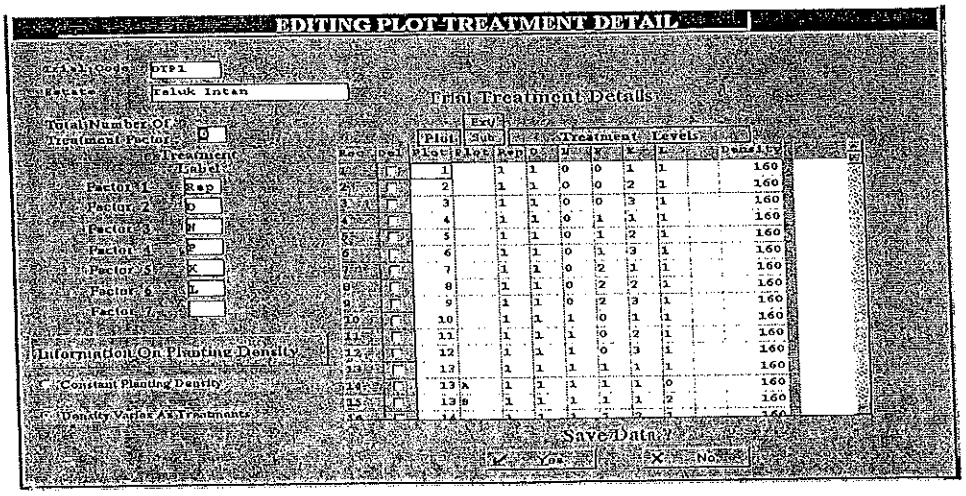
5) Click *'Check For Trial Code Availability'* button.

6) Click *'Start Replacing Data From Backup File'* to start the reading process or otherwise, click *'Exit Read Backup'*.

   - Your computer screen would look like Figure 9.

7) Save the data record by clicking *'Yes'* button or ignore it by clicking *'No'* button.

Figure 9: Read Treatment Backup File (Editing Plot Treatment Detail)



2.9 Reading Treatment From 'non-ReaDA' DBase File

This procedure is used when we need to read the data that has been saved as a database file by using the 'non-ReaDA' system. However, if the raw data is in other spreadsheet (e.g. Excel, Quattro Pro etc.), you need to convert the file as a *.dbf first. While reading the file, you can also edit the data.

The following steps are to be taken:

1) Click *'Read Treatment From 'Non-ReaDA' DBase File'* button and your screen will show *'Read Treatment From 'Non-ReaDA' DBase File'* form.

2) *Select Code To Be Read.* e.g. DTP1 Teluk Intan

3) Click *'Yes/Continue'* button to continue the procedure.

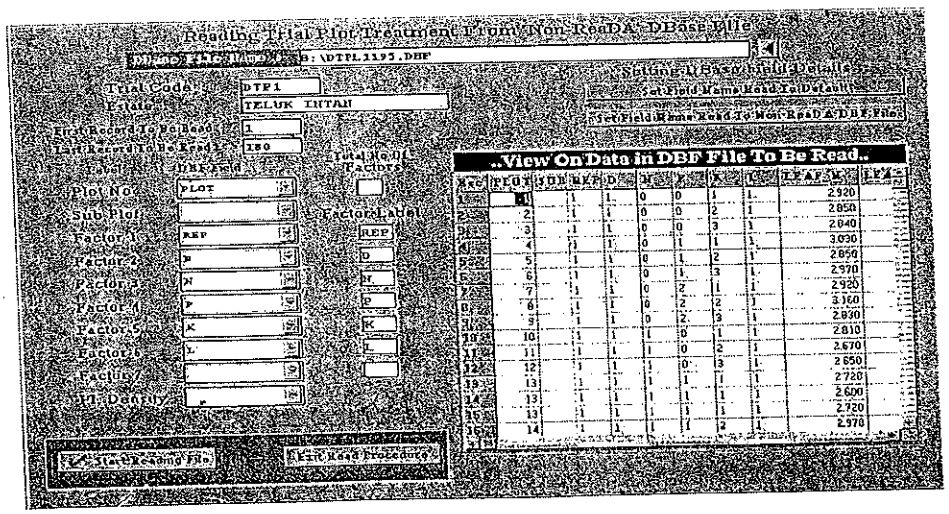
- *'Reading Trial Plot Treatment From 'Non-ReaDa Dbase File'* form will appear.

4) Enter the external *Dbase File Name:* by keying in the blank column provided or selecting from the listed file name. This is an external database file.

e.g. B:\DTPL1195.DBF

- The screen will show *'Reading Trial Plot Treatment From 'Non-ReaDa' Dbase'* with the *'View On Data in DBF File To Be Read'* form like Figure 10.

Figure 10: Reading Trial Plot Treatment From 'Non-ReaDa' Dbase File



5) *First Record To Be Read:* State the first record you want to read according to external database file.

6) *Last Record To Be Read:* State the last record to read.

7) Select your requirement for *'Setting Details Dbase Field'*.

- *'Set Field Name Read To Default:* When 'non-ReaDA' database field name is similar to ReaDA field name.
- *'Set Field Name Read To Non-ReaDA DBF File':* When 'non-ReaDA' database field name differs from your ReaDA field name. You can enter a new field name or select from the list under the *'DBF Field'* column.

8) Fill in the *'DBF Field'* column and *'Factor Label'*.

9) Click *'Start Reading File'* button and your screen will show *'View and Edit Read Data'*.

- Any editing of the data record can be done here.

10) '*Save'* your data records or *'Cancel'*.

2.10 Deleting Trial Plot Treatment

When you confirm your request to delete, you may delete your data records permanently.

The steps are as follows:

1) Click *'Delete Trial Plot Treatment'* button and your screen will show *'Deleting Trial Treatment'* form.

2) *Select Code To Be Deleted*

e.g. DTP1 Teluk Intan

3) Click *'Yes/Continue'* button or otherwise click *'Cancel-Exit'*.

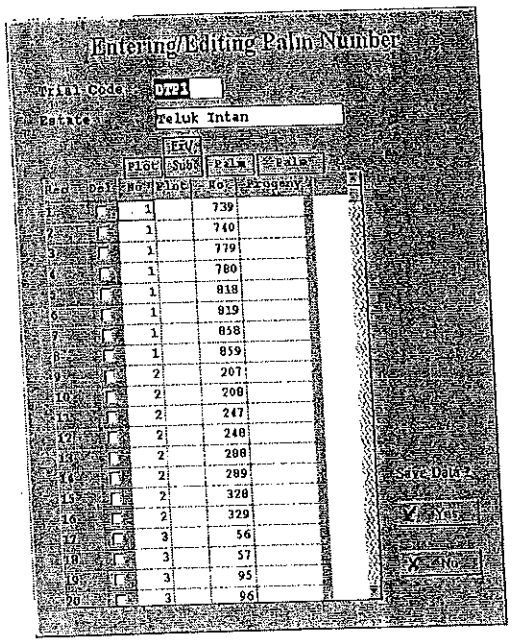
- To continue the deleting procedure, you should ensure that all the data record for DTP1 has already been deleted.

4) Click '*Yes'* button to confirm your deletion or click '*Cancel'* to quit without deleting the selected file.

2.11 Entering /Editing Palm Numbering

You can enter or edit a new data for palm numbering in the same form and procedures.

Figure 11: Entering/Editing Palm Numbering



Steps:

1) Click *'Entry/Edit On Palm Numbering'* button and the screen will show *'Entry/Editing Palm Numbering'* .

2) *Select Code To Be Edited* by clicking the mouse.

e.g. DTP1 Teluk Intan

3) Click *'Yes/Continue'* button to continue the procedure or *'Cancel-Exit'* to quit.

- Your screen would look like Figure 11.

4) Start entering /editing the data where needed.

5) Save your entry/edited data by clicking *'Yes'* button, or otherwise click *'No'*.

2.12 Saving Palm Numbering As a Database File.

This procedure is used when you need to save your data records for palm numbering.

Steps:

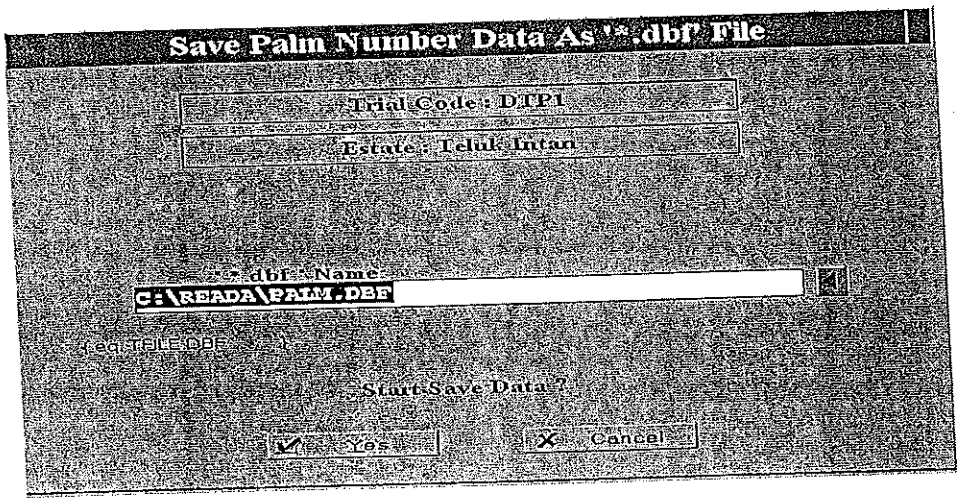
1) Click *'Save Palm Number As *. dbf File'* button and *'Save Palm Number As *. dbf File'* will appear.

2) *Select Code To Be Saved* by clicking your mouse to the selection.

3) Click *'Yes/Continue'* button, or otherwise click *'Cancel-Exit'.*

- Your screen will show *'Save Palm Number Data As*.dbf File'* as in Figure 12.

Figure 12: Save Palm Number Data As *.dbf File



4) Enter your *'*.dbf' Name:* in the column provided or browse the directories and select the file name by double-clicking to the selection.

5) Click *'Yes'* button to save your data record or click *'Cancel'* to quit the procedure without saving it.

2.13 Reading Palm Numbering From Back Up File

This procedure is used when you need to recover the data record for palm numbering especially when you have deleted a particular file.

Steps:

1) Click '*Read Palm Number Back Up File'* button and your screen will show *'Read Palm Number From Back Up File'* form.

2) *Select Code To Be Read* by clicking the selection.

e.g. DTP1 Teluk Intan

3) Confirm your selection by clicking *'Yes/Continue'* button or '*Cancel-Exit'* to quit the procedure.

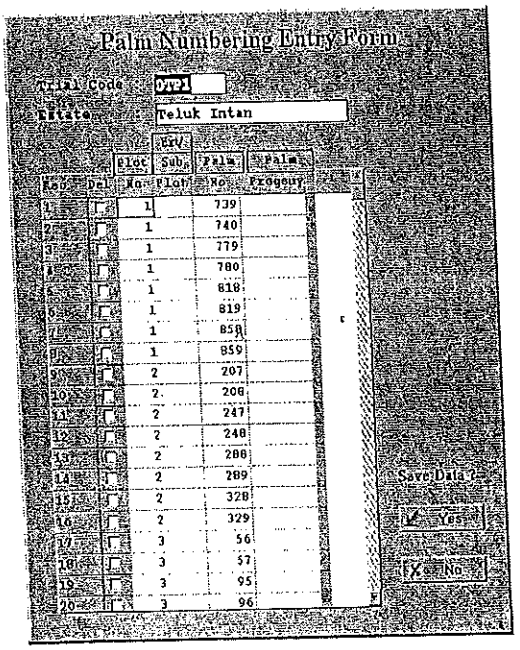
4) Enter the *Back Up File Name* with its drive and directory or browse the templates.

5) Click *'Check For Trial Code Availability'* button.

6) *Select Code To Be Read* from the list.

7) Click *'Start Replacing Data From Back Up File'* button and the screen will look like Figure 13.

Figure 13: Palm Numbering Entry Form.



8) Save your data by clicking *'Yes'* button or otherwise *'No'*.

2.14 Reading Palm Number From 'non-ReaDA' DBase file.

The procedure of 'reading data record from 'non-ReaDA' database file' is used when we need to read the data that was saved as a database file by using the 'non-ReaDA' system. While reading the file, you can also edit the data.

Steps:

1) Click *'Read Palm Number From 'Non-ReaDA' Dbase File'* button.

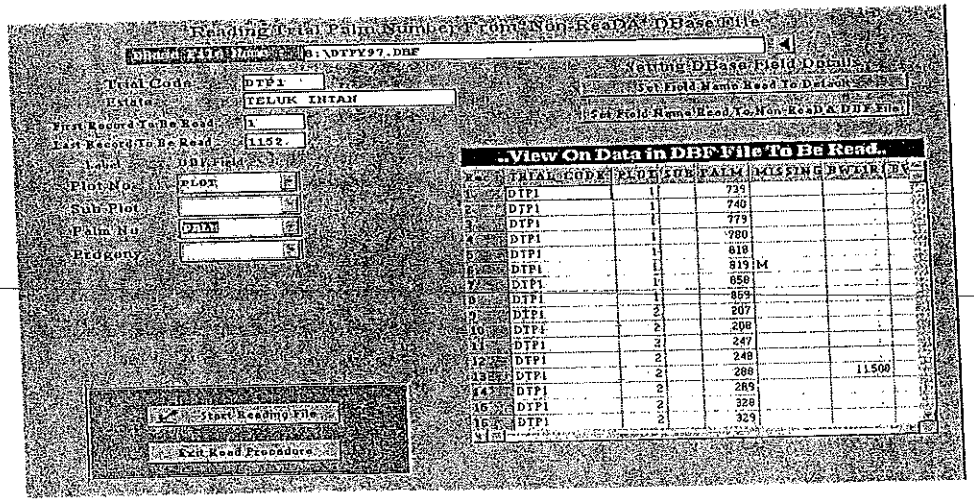
2) *Select Code To Be Read.* e.g. DTP1 Teluk Intan

3) Click *'Yes/Continue'* button to continue the procedure or *'Cancel-Exit'* to ignore it.

4) Enter your external *Dbase File Name:* e.g. B:\DTPY97.DBF

- Your screen would look like Figure 14.

Figure 14: Reading Trial Palm Number From 'Non-ReaDA' Dbase File



5) Select for *'Setting Dbase Field Details'*.

- Click *'Set Field Name Read To Default'* button if the field name in 'non-ReaDA'*.dbf file is the same as your field name in 'Non-ReaDA'*.dbf file.
- Click *'Set Field Name Read To Non-ReaDA DBF File'* button if the field name in 'non-ReaDA'*.dbf file is different from your field name in 'ReaDA'*.dbf file.

Then you need to fill in the *'DBF Field'* column according to *'View On Data In DBF File To Be Read'.*

6) *First Record To Be Read:* State the first record you want to read.

7) *Last Record To Be Read:* State the last record you want to be read.

8) Click *'Start Reading File'* button and *'View And Edit Read Data'* will appear in the form.

- Any editing can be done here.

9) Click *'Save'* button to save the data records otherwise click *'Cancel'* button.

2.15 Deleting Plot Palm Numbering

This procedure is used when you need to delete the record for plot palm numbering.

Steps:

1) Click *'Delete Plot Palm Numbering'* button and *'Deleting Trial Palm Numbering'* will appear.

2) *Select Code To Be Deleted.*

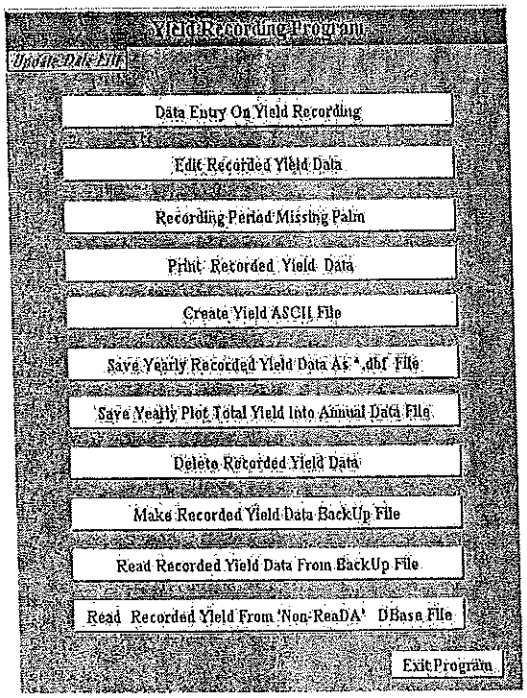
3) Click *'Yes/Continue'* button to continue the deleting procedure or click *'Cancel-Exit'* to quit the procedure.

4) Click *'Yes'* button to confirm your deletion or otherwise click *'Cancel'*.

Lesson 3

YIELD RECORDING PROGRAM

Figure 15: Yield Recording Program

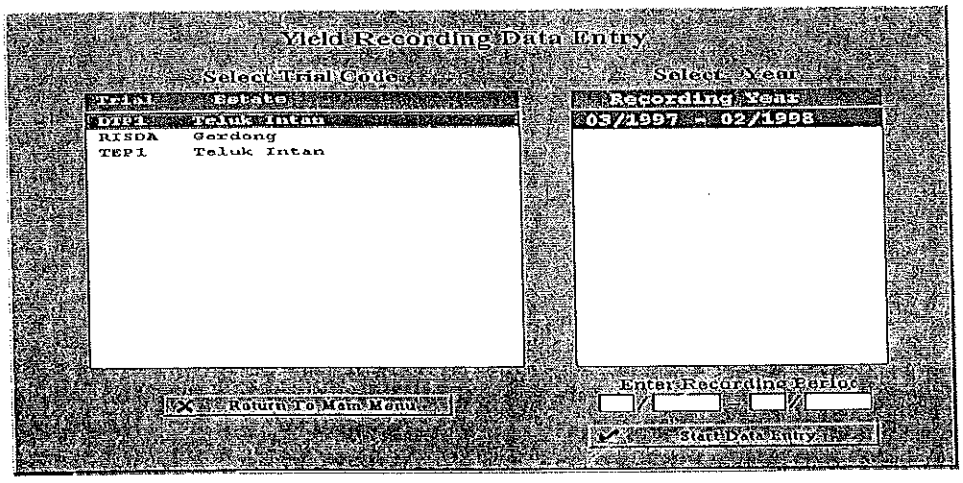


3.1 Enter New Yield Data

Steps:

1) Click *'Data Entry On Yield Recording'* button and your screen will show the *'Yield Recording Data Entry'* as in Figure 16.

Figure 16: Yield Recording Data Entry



- Code selection form will appears for every procedure as in Figure 16.

2) ***Select Trial Code:*** Select the particular code by clicking your mouse according to your selected estate name.

e.g. DTP1 Teluk Intan

3) ***Select Year:*** Select the recording year by clicking the mouse to the selection or enter a new recording period into the box provided.

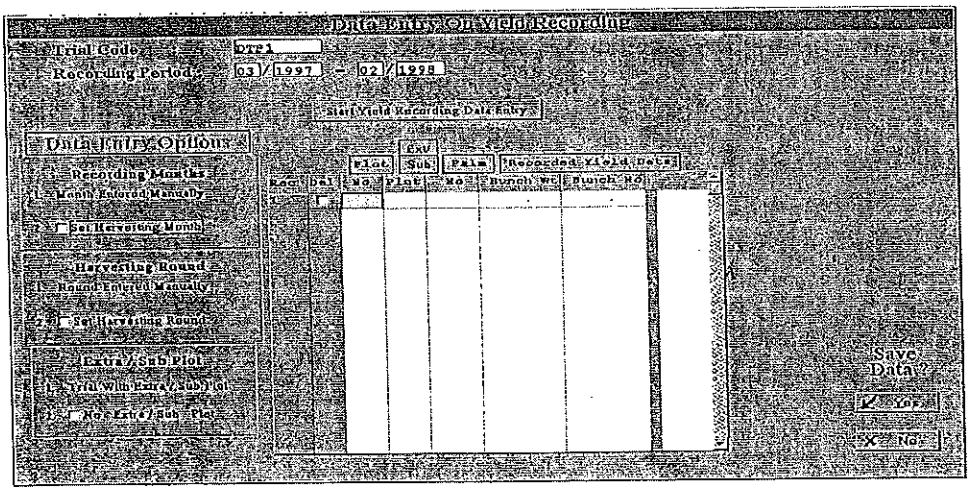
e.g. 03/1997-02/1998

4) Click *'Start Data Entry'* button, otherwise *'Return To Main Menu'.*

5) ***Data Entry Option:*** Select the option for the data entries.

- ***Set Harvesting Month:*** If you have decided on the particular harvesting month, then enter the number into the box beside.
- ***Set Harvesting Round:*** If you have decided on the particular harvesting round for data entry, then enter the number into the box beside.
- *No Extra/Sub Plot.*

Figure 17: Data Entry On Yield Recording

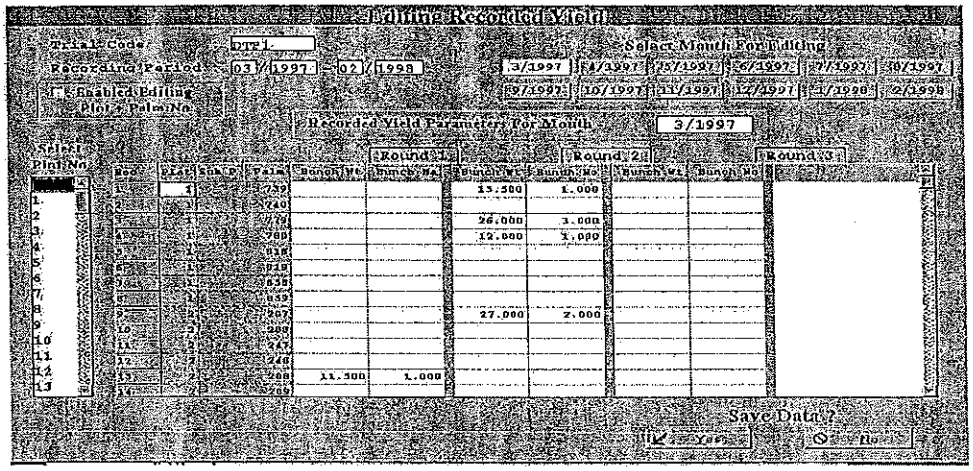


6) Click *'Start Yield Recording Data Entry'* button.

- Your screen would look like Figure 17.
- Start entering the data.

3.2 Editing The Yield Data Record

Figure 18: Editing Recorded Yield



Steps:

1) Click ***'Edit Recorded Yield Data'*** button.

- ***'Edit Recorded Yield Data'*** appears on the screen.

2) ***Select Trial Code:*** Click your mouse after a selection is made.

3) ***Select Year:*** Click 'data recording years' you want to edit.

4) Click ***'Continue With Selection'*** button if you want to continue with the procedure or otherwise click ***'Return to Main Menu'***.

- Your screen will look like Figure 18.
  (Editing Recorded Yield)

5) ***Select Month For Editing:*** Click the required month.

6) ***Enabled Editing Plot + Palm No:*** This is used when you decide to change the plot or palm number.

7) ***Select Plot No:*** Click the plot number to find the palm number.

3.3 Recording Period Missing Palm

This form is used to record any missing palm and to view all the missing palms that have been recorded.

Steps:

1) Click *'Recording Period Missing Palm'* button and the *'Mark Missing Palm'* will appear.

2) *Select Trial Code*: Click the selected trial code.

e.g. DTP1 Teluk Intan.

3) *Select Year:*

4) Click *'Continue With Selection'* button to further the next steps or otherwise, click *'Return to Main Menu'.*

- If you continue, your screen will show the *'Trial Plot Missing Palm'.*

5) Select either *'Enter Missing Palm Number'* or *'Mark Missing Palm'* button.

- *'Enter Missing Palm Number'* when you need to state the missing palm number.

  Your screen would look like Figure 19.1.

- *'Mark Missing Palm'* when you need to view the listed missing palm number. You can also mark the missing palm number by entering the character 'M' into the column. You can select the plot number to find the palm number easily by clicking the desired *Plot No.*

  Your screen would look like Figure 19.2.

Figure 19.1:Trial Plot Missing Palm (Enter Missing Palm Number)

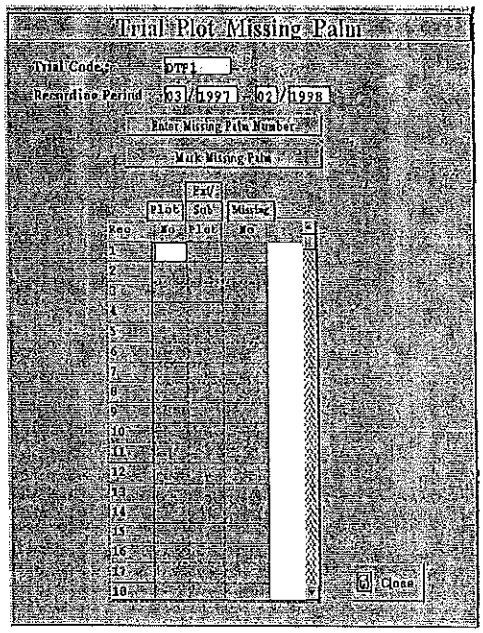
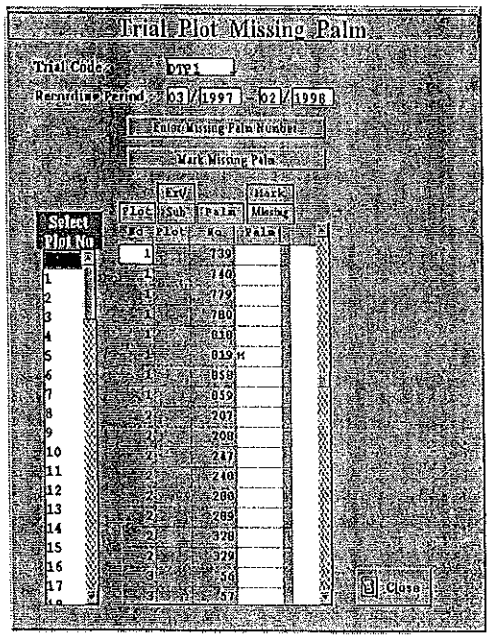


Figure 19.2:Trial Plot Missing Palm (Mark Missing Palm)



6) Click *'Close'* button to return to the main menu and the program will automatically save the data to file.

3.4 Printing Yield Data Record

The 'Print Recorded Yield Data' is used when you want to print out the yield data record.

Steps:

1) Click *'Print Recorded Yield Data'* button.

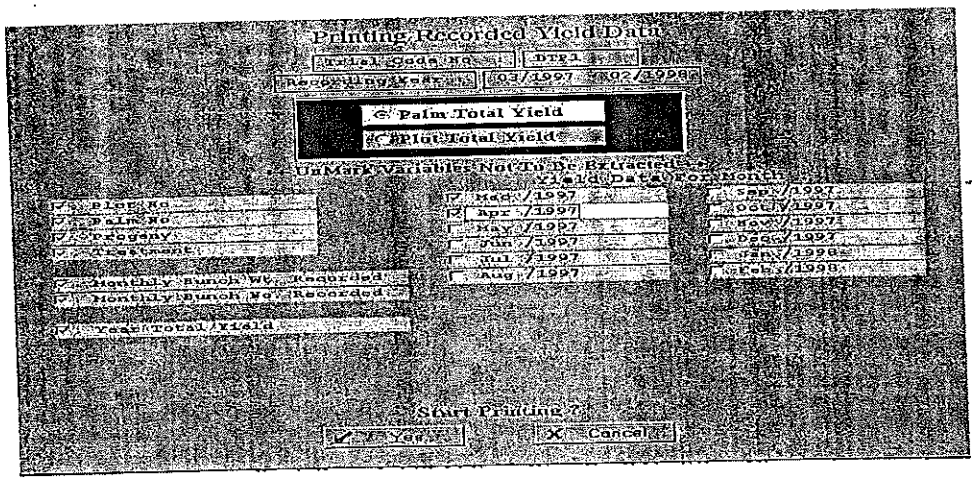
2) *Select Trial Code*: Click desired trial and estate name.

3) *Select Year:* Click the particular year recorded in yield data.

4) Click *'Continue With Selection'* button or otherwise click *'Return to Main Menu'.*

- If you continue, your screen will show the *'Printing Recorded Yield Data'* as in Figure 20.

Figure 20: Printing Recorded Yield Data



5) Select either *'Palm Total Yield'* or *'Plot Total Yield'* for printing.

- Mark the field name needed.
- Example of printed report in Appendix 1(a) for 'Palm Total Yield' and Appendix 1(b) for 'Plot Total Yield'.

6) *Yield Data For Month:* Mark the selected month needed.

3.5 Creating Yield Data Record as ASCII File

This procedure is important as a prerequisite for running data in any computer program (esp. SAS) for making data analysis.

Steps:

1) Click *'Create Yield ASCII File'* button.

2) *Select Trial Code*: e.g. DTP1 Teluk Intan

3) *Select Year*: e.g. 03/1997 - 02/1998

4) Click *'Continue With Selection'* button or otherwise click *'Return To Main Menu'*.

- Your screen will look like Figure 21.

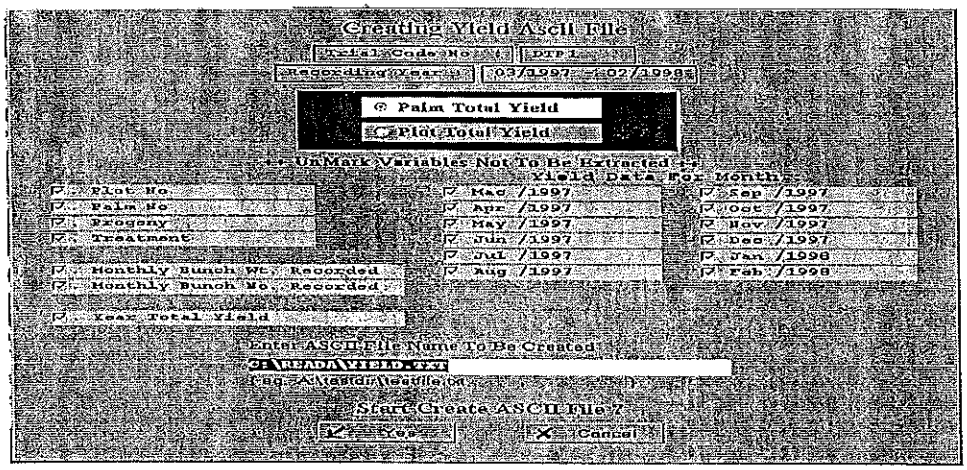
(Creating Yield ASCII File)

5) Enter the *ASCII File Name*.

e.g. C:\READA\YIELD.TXT

6) *Start Create ASCII File* by clicking *'Yes'* button or otherwise click *'Cancel'* button.

Figure 21: Creating Yield ASCII File



3.6 Saving Recorded Yield Data To Database File.

Steps:

1) Click *'Save Yearly Recorded Yield Data As *.dbf File'.*

- *'Save Yearly Recorded Yield Data As *.dbf File'* will appear.

2) *Select Trial Code:* e.g. DTP1 Teluk Intan

3) *Select Year :* e.g. 03/1997 - 02/1998

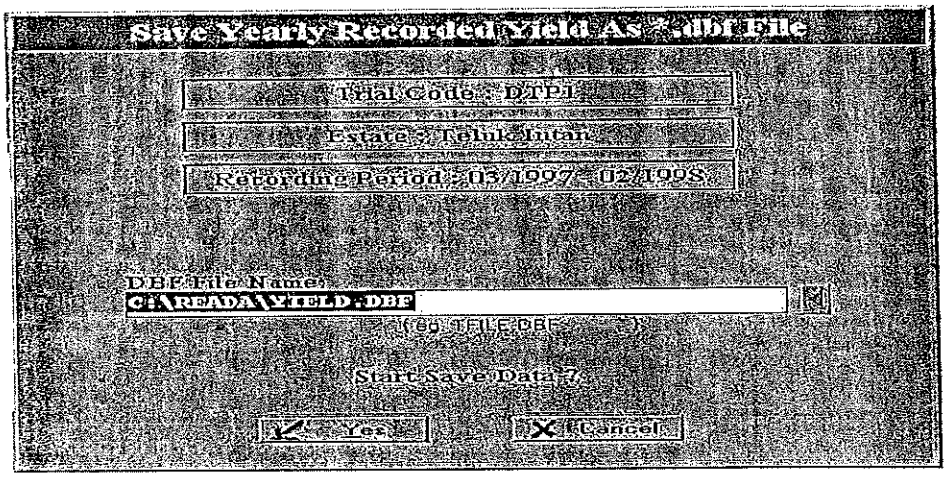
4) Click *'Continue With Selection'* button if you want to continue the saving procedure or click *'Return to Main Menu'* button if you want to quit.

- Now, your screen would look like Figure 22.

5) Enter the *DBF File Name* (Drive/Directory/Filename.dbf)

e.g. C:\READA\YIELD.DBF

Figure 22: Save Yearly Recorded Yield As *.dbf File



6) Click *'Yes'* button to start the saving process or click *'Cancel'* button if you want to ignore it.

3.7 Saving Yearly Plot Total Yield Into Annual Data File

This procedure is used when we need to save the summary of the calculated yield/tonne/hectare, Bunch no./palm/year, mean bunch weight and Plot Total Yield/palm/year.

Steps:

1) Click *'Save Yearly Plot Total Yield Into Annual Data File'* button.

2) *Select Trial Code*: e.g. DTP1 Teluk Intan

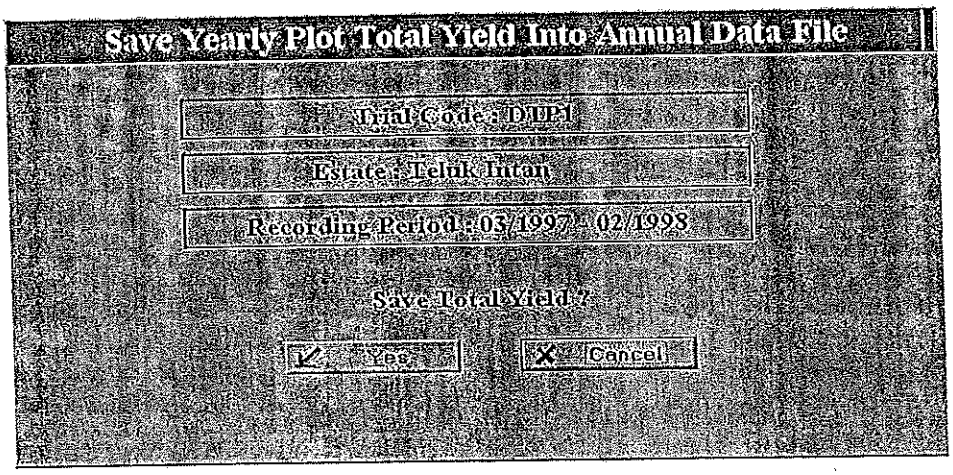
3) *Select Year*: e.g. 03/1997 - 02/1998

4) Click *'Continue With Selection'* button to go to the next procedure or click *'Return To Main Menu'* button to quit from the process.

- Your screen would look like Figure 23.

  (Save Yearly Plot Total Yield Into Annual Data File)

Figure 23: Save Yearly Plot Total Yield Into Annual Data File



5) Confirm your selection by clicking *'Yes'* button or otherwise click *'Cancel'* button.

3.8 Deleting The Trial Data Record

By using this procedure, it may delete your one year of data entry permanently.

Steps:

1) Click *'Delete Recorded Yield Data'* button.
2) *Select Trial Code:* e.g. DTP1 Teluk Intan
3) *Select Year:* e.g. 03/1997 - 02/1998
4) Click *'Return To Main Menu'* button to quit the deleting procedure or click *'Continue With Selection'* button if you want to proceed with it.
5) Confirm your deletion by clicking *'Yes'* button and your data record will be deleted permanently or otherwise *'Cancel'* button to ignore it.

3.9 Making the Backup File for Trial Yield Data

This procedure is important to prevent you from losing the file record permanently even though you have deleted the file before.

Steps:

1) Click *'Make Recorded Yield Data Back Up File'* button.

2) *Select Trial Code:* e.g. DTP1 Teluk Intan

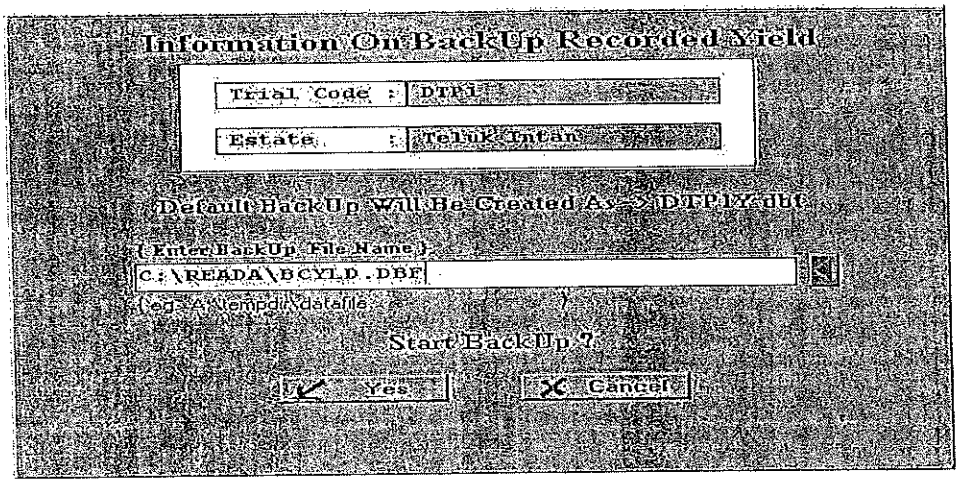
3) Click *'Return to Main Menu'* button if you want to quit the procedure or click *'Continue With Selection'* button if you want to continue with the procedure.

- Your screen will show *'Information On BackUp Recording Yield'* as in Figure 24.

4) Enter the *Back Up File Name*

e.g. C:\READA\BCYLD.DBF

Figure 24: Information On Back Up Recorded Yield



5) Click *'Yes'* button if you want to start making the back up file or otherwise click *'Cancel'* button.

3.10 Reading Yield Data From Back Up File.

This procedure is used when you need to recover the data record that has been deleted before. This process can be done if you have created the back up file for the data.

Steps:

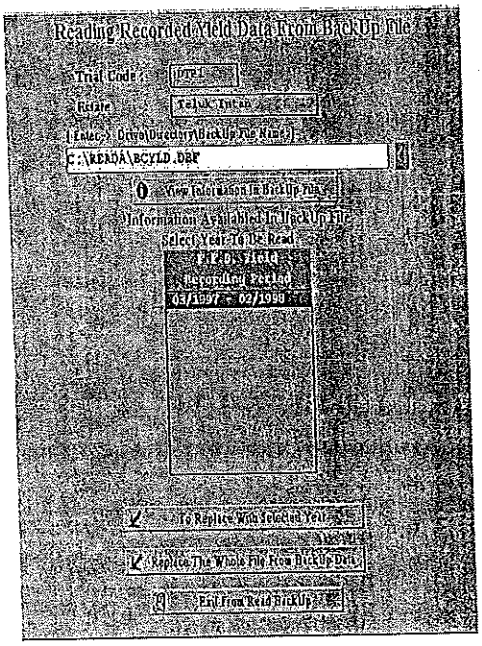
1) Click *'Read Recorded Yield Data From Back Up File'* button.

2) *Select Trial Code:* e.g. DTP1 Teluk Intan

3) Click *'Continue With Selection'* button to continue with the procedure or otherwise click *'Return To Main Menu'* button.

- If you continue, your screen would look like Figure 25.

Figure 25: Reading Recorded Yield Data From Backup File



4) Enter *BackUp File Name*

- Or you can browse the directories by clicking  toolbar and search for a particular file. Select the file by double-clicking your mouse.

5) Click *'View Information In Back Up File'* button and you can see the year listed.

- Select the recorded year.

6) Click *'To Replace With The Selected Year'* button if you want to read and replace the data for a specific year only, or otherwise click *'Replace The Whole File From Back Up Data'* button to read and replace the whole file. Click *'Exit From Read Back Up'* button if you want to quit from the reading process.

3.11 Reading Yield Record from 'non-ReaDA' Dbase File.

This procedure is used when we need to read the data record that has been saved as a database file by using the 'non-ReaDA' system. However, if you have saved the raw data used the other spreadsheet, you should convert it to database file first. While reading the file, you can also edit the data.

Steps:

1) Click *'Read Recorded Yield From'Non-ReaDA' Dbase File'* button.

2) *Select Trial Code:* e.g. DTP1 Teluk Intan

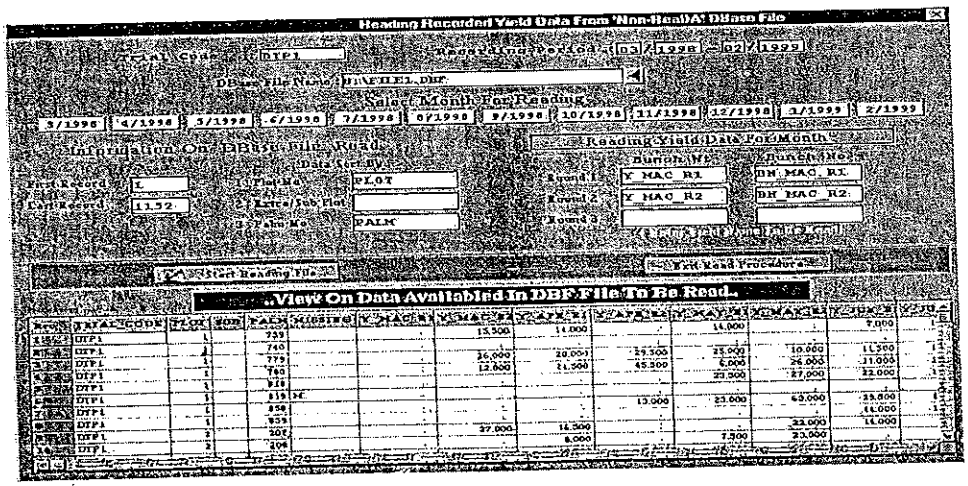
3) *Select Year* by clicking the mouse under the *'Recorded Year'* column or enter the recording period into the box provided.

e.g. 03/1998 - 02/1999

4) Click *'Continue With Selection'.*

- Your screen will show the *'Reading Recorded Yield Data From 'Non-ReaDA' DBase File* as in Figure 26.

Figure 26: Reading Recorded Yield Data From Non-ReaDA DBase File



5) Enter the external ***Dbase File Name*** or browse the directory.

e.g. B:\FILE1.DBF

6) ***Select Month For Reading***. e.g. 03/1998

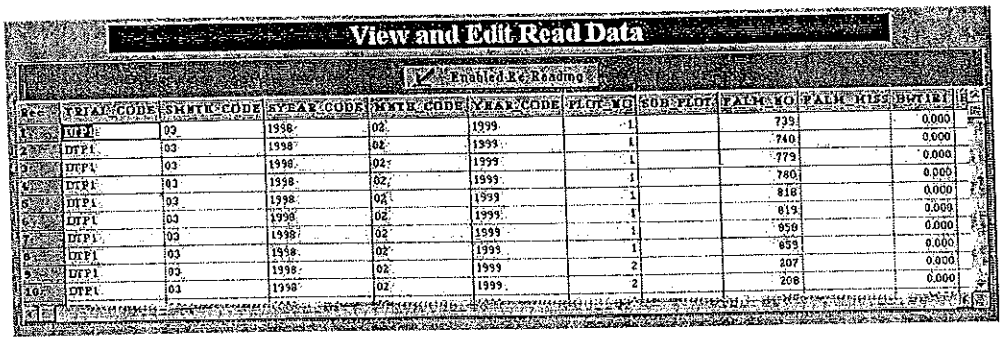
7) Enter the ***First Record*** and ***Last Record*** to be read.

8) Enter ***Data Sort By*** accordingly to ***Information On Dbase File Read*** such as ***Plot No, Extra/Sub Plot, Palm No*** and ***Bunch Wt*** and ***Bunch No*** of ***Round.***

9) Click ***'Starting to Read File'*** to start the reading procedure.

- The ***'View and Edit Read Data'*** form as in Figure 26.1 will appear. You can edit as well.

Figure 26.1:



10) Click ***'Enabled Re-reading'*** button in order to go back to ***'Reading Recorded Yield Data From 'Non-ReaDA' DBF File'*** form.

11) Click ***'Exit Reading Procedure'*** button.

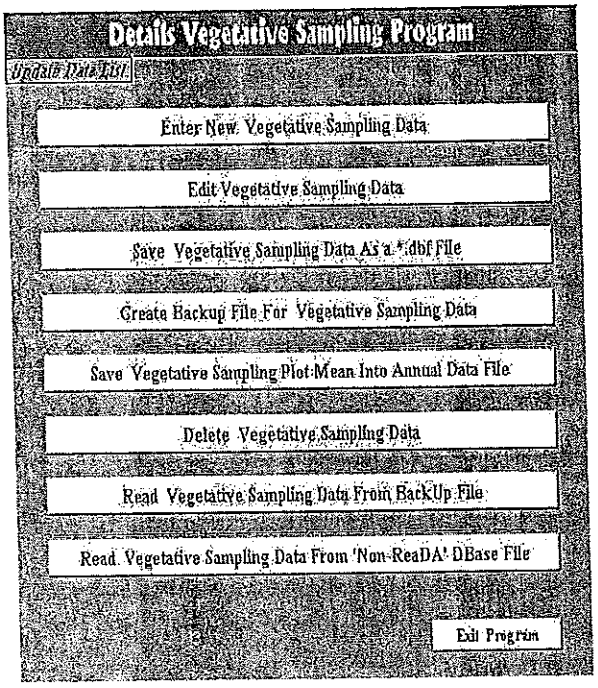
12) Save your data records.

AHMAD AFANDI B. MURDI
Pegawai Penyelidik
Stesen MPOB Lahad Datu

Lesson 4

VEGETATIVE FIELD SAMPLING

Figure 27: Details Vegetative Measurement Program



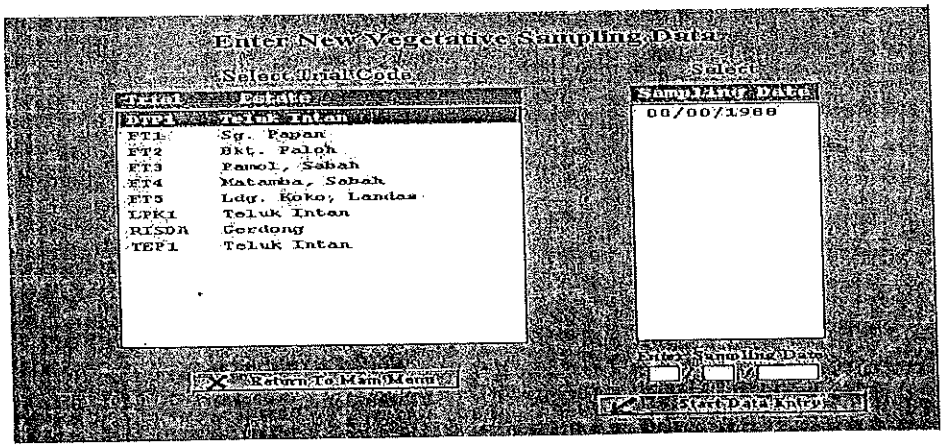
4.1 Entering New Vegetative Sampling Data

This procedure is used to create a new vegetative data record.

The steps are as follows:

1) Click *'Entering New Vegetative Sampling Data'* button and *'Entering New Vegetative Sampling Data'* form appears as in Figure 28.

Figure 28: Entering New Vegetative Sampling Data



- Code selection form appear for every procedure as in Figure 28.

2) *Select Trial Code:* e.g. DTP1 Teluk Intan

3) *Select Sampling Date* or you can enter a new sampling date into the boxes provided.

e.g. 00/00/1988

4) Click *'Start Data Entry'* button.

5) Select either *'Main Vegetative Measurement'* or *'Leaflets & Rachis Measurement'* by clicking the tab box at the bottom of the form.

- *Main Vegetative Measurement:* The variables to be used are Palm Girth (cm), Palm Height (cm), Fronds Production (No.), No. of Fronds (No.), and Frond Length (cm).
- *Leaflets & Rachis Measurement:* The variables to be used are No of Pinnae (No.), Leaflets Value (cm), Rachis Width (cm), and Rachis Depth (cm).

6) Start entering data.

7) Save your record by clicking *'Yes'* button otherwise click *'No'* button.

4.2 Editing Recorded Vegetative Sampling Data

This form is used when you need to edit the vegetative data records.

The steps are as follow:

1) Click *'Edit Vegetative Sampling Data'* button and you can see the *'Edit Vegetative Sampling Data'* form.

2) *Select Trial Code :* e.g. DTP1 Teluk Intan

3) *Select Sampling Date* by clicking the selected date.

e.g. 00/00/1988

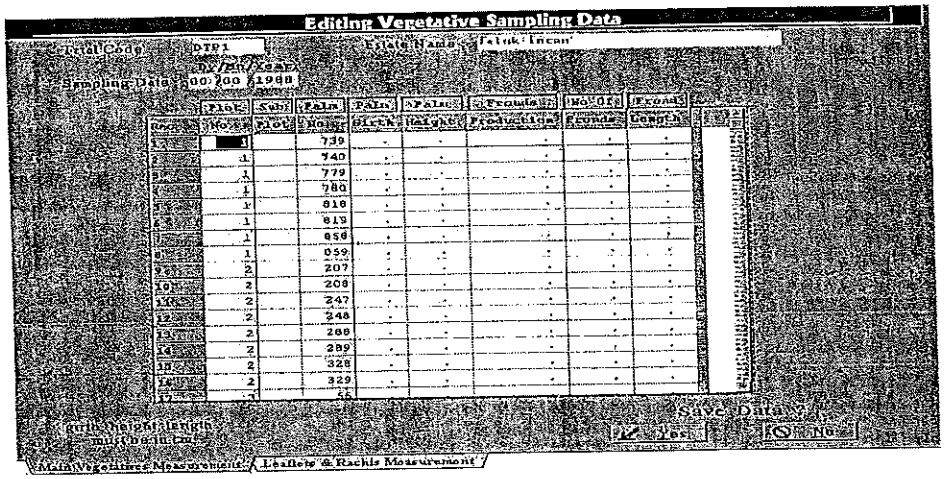
4) Click *'Continue With Selection'* button to go to the next step or otherwise, click *'Return To Main Menu'* button.

- *'Editing Vegetative Sampling Data'* appears.

5) Select the form required, either *'Main Vegetative Measurement'* or *'Leaflets & Rachis Measurement'*.

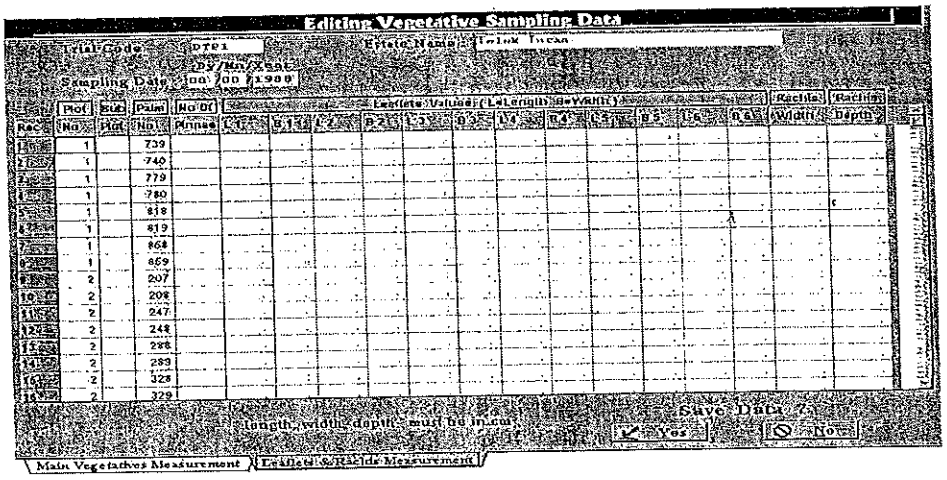
- Your screen, if you select *'Main Vegetative Measurement'*, will look like Figure 29.1.

Figure 29.1: Editing Vegetative Sampling Data (Main Vegetative Measurement)



- Your screen, if you select *'Leaflets & Rachis Measurement'*, will look like Figure 29.2.

Figure 29.2: Editing Vegetative Sampling Data (Leaflets & Rachis Measurement)



6) Click *'Yes'* button to save the records.

4.3 Saving Vegetative Sampling Data as Database File.

This procedure is used when we need to save the vegetative data record as an external database using the ReaDA system format.

Steps:

1) Click *'Save Vegetative Sampling Data As *.dbf File'* button.

2) *Select Trial Code:* Click your selected trial code and estate name.

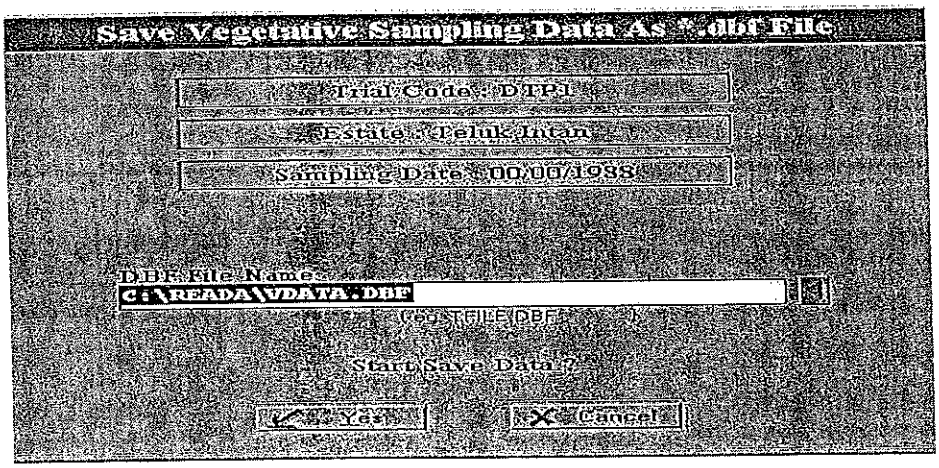
e.g. DTP1 Teluk Intan

3) *Select Sampling Date:* e.g. 00/00/1988

4) Click *'Continue With Selection'* button to continue with the procedure or click *'Return To Main menu'* button to quit.

- If you continue, your screen would look like Figure 30.

Figure 30: Save Vegetative Sampling Data As *.dbf File



5) Enter *DBF File Name* with its drive and the directory if necessary.

e.g. C:\READA\VDATA.DBF

6) Confirm saving the data by clicking *'Yes'* button otherwise click *'Cancel'* button.

4.4 Creating Backup File For Vegetative Sampling Data

The following steps need to be taken in order to provide the backup file for the data records:

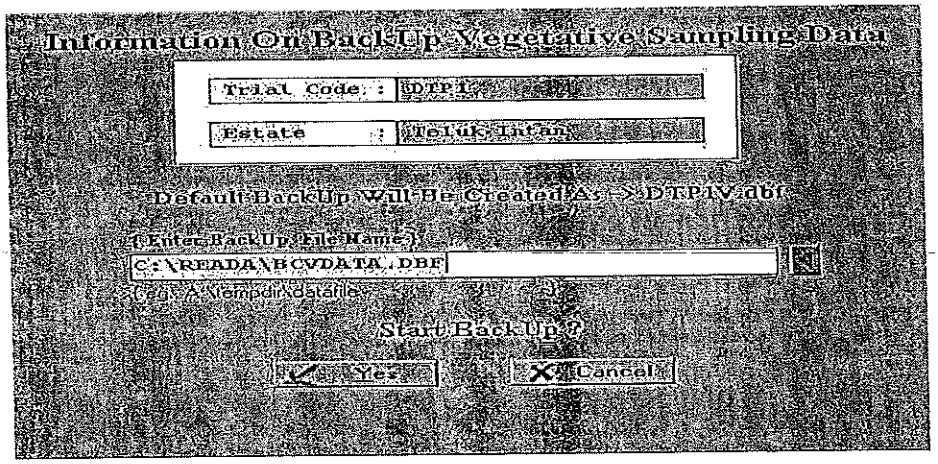
1) Click *'Create Backup File For Vegetative Sampling Data'* button.

2) *Select Trial Code:* e.g. DTP1 Teluk Intan

3) Click *'Continue With Selection'* button to continue with the procedure otherwise click *'Return To Main Menu'* button.

- If you continue, your screen would look like Figure 31.

Figure 31: Information On BackUp Vegetative Sampling Data



4) Enter your *Back Up File Name* (Drive/Directory/Filename.dbf)

e.g. C:\READA\BCVDATA.DBF

5) Confirm your selection by clicking *'Yes'* button or click *'Cancel'* button if you want to exit from the procedure.

4.5 Saving Vegetative Sampling Plot Mean Into Annual Data File

This procedure is used to calculate plot value on all the variables of the vegetative measurement data especially for the frond area and the frond dry weight that require additional calculation. The calculated data will be saved into annual data file.

Steps:

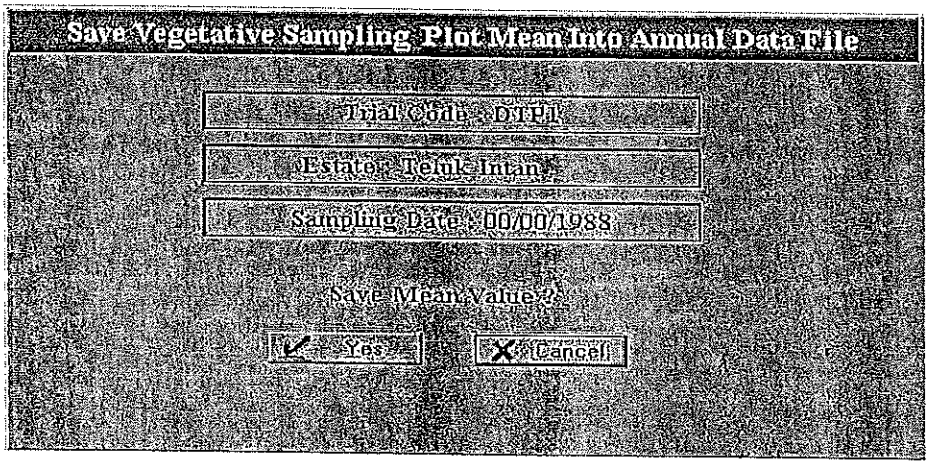
1) Click *'Save Vegetative Sampling Plot Mean Into Annual Data File'* button and *'Save Vegetative Sampling Plot Mean Into Annual Data File'* will appear.

2) *Select Trial Code:*

3) Click *'Continue With Selection'* button otherwise click *'Return To Main Menu'* button.

- *'Save Vegetative Sampling Plot Mean Into Annual Data File'* that appears will look like Figure 32.

Figure 32: Save Vegetative Sampling Plot Mean Into Annual Data File.



4) Save mean value by clicking *'Yes'* button otherwise click *'Cancel'* button.

4.6 Deleting Vegetative Sampling Data

Steps:

1) Click *'Delete Vegetative Sampling Data'* button.

2) *Select Trial Code.*

3) *Select Sampling Date.*

4) Click *'Continue With Selection'* button otherwise click *'Return To Main Menu'* button.

- If you decide to continue, the *'Information On Vegetative Sampling Data To Be Deleted'* will appear.

5) Confirm by clicking *'Yes'* button and your data record will be deleted permanently, otherwise click *'Cancel'* button.

4.7 Reading Vegetative Sampling Data From Back Up File

This procedure is used when you need to retrieve the records that have been deleted before.

Steps:

1) Click *'Read Vegetative Sampling Data From Back Up File'* button and *'Read Vegetative Sampling Back Up File'* will appear on the screen.

2) *Select Trial Code:* DTP1 Teluk Intan

3) Click *'Continue With Selection'* button to proceed with the procedure or click *'Return To Main Menu'* button to quit it.

- *'Reading Vegetative Sampling Data From Back Up File'* will appear.

4) Enter the *Back Up File Name* with its drive and the directory.

e.g. C:\READA\BCVDATA.BDF

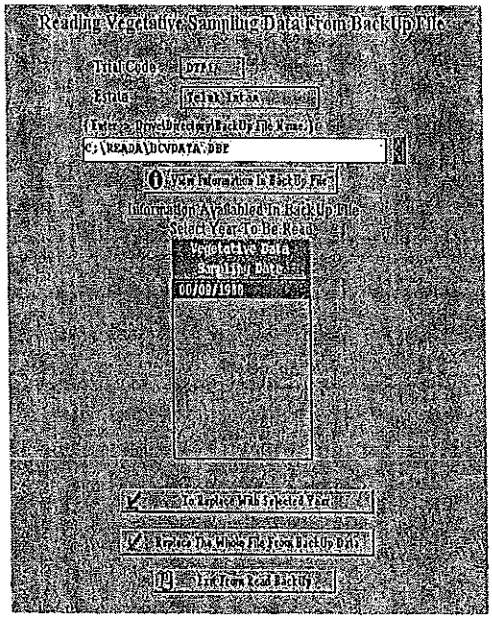
- You can also browse the directory and select a particular file.

5) Click *'View Information In Back Up File'.*

- *'Vegetative Data Sampling Date'* will appear which contains the year.

- Your screen would look like Figure 33.

Figure 33: Reading Vegetative Sampling Data From BackUp File



6) *Select Year To Be Read.*

7) Select either *'To Replace With Selected Year'*, *'Replace The Whole File From Back Up Data'* or *'Exit From Read BackUp'.*

4.8 Reading Vegetative Data From 'non-ReaDA' DBase File.

The process of reading vegetative data record from 'non-ReaDA' DBase file is used when you need to read the data that has been saved in a database file by using the 'non-ReaDA' system (external database file).

Steps:

1) Click *'Read Vegetative Sampling Data From 'Non-ReaDA' DBase File'* button.

- *'Read Vegetative Sampling Data From 'Non-ReaDA' Dbase File'* will appear.

2) *Select Trial Code.*

3) *Select Sampling Date* or enter a new sampling date into the box provided. e.g. 00/00/1988

4) Click *'Start Data Entry'* button.

- Your screen would look like Figure 34 with *'Reading Vegetative Sampling Data From 'Non-ReaDA' DBase File'* appearing.

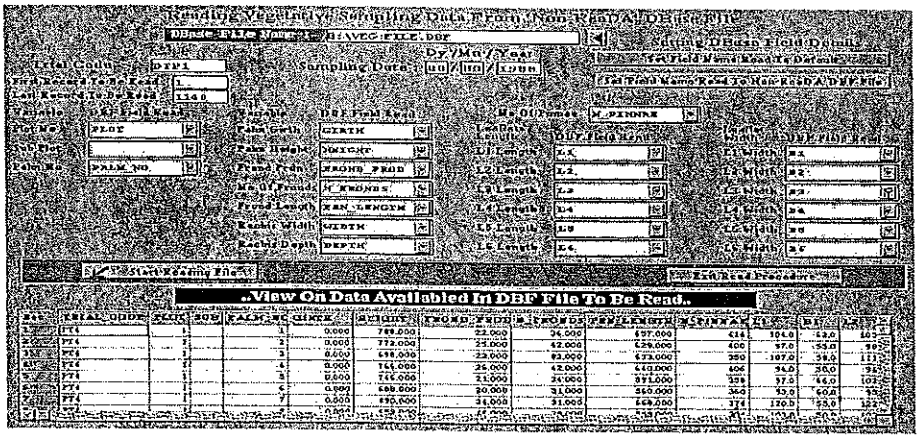
5) Enter the external *DBase File Name* or browse the directory.

e.g. B:\VEG_FILE.DBF

6) *'Setting DBase Field Details'* between the external database file and the ReaDA database file.

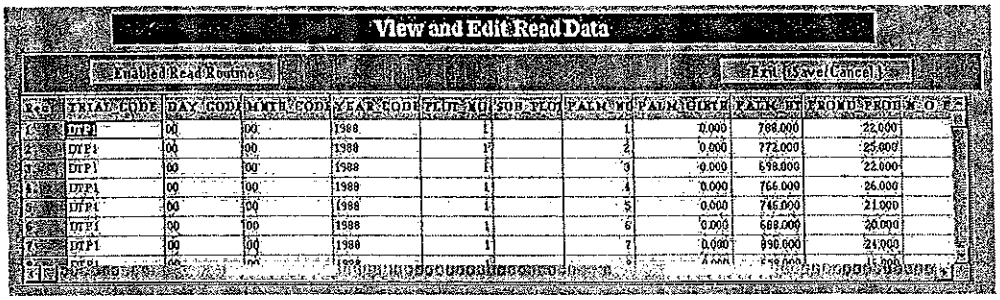
- *Set Field Name Read To Default*
- *Set Field Name Read To Non-ReaDA DBF File:* You will have to set up the *DBF Field Read* accordingly to external database field name.

Figure 34: Reading Vegetative Sampling Data From 'Non-ReaDA' DBase file



7) Click *'Start Reading File'* button and your screen will show *'View and Edit Read Data'* as in Figure 34.1, otherwise click *'Exit Reading Procedure'* button to return to the main menu.

Figure 34.1: View and Edit Read Data



Rec	TRIAL CODE	DAY CODE	MTH CODE	YEAR CODE	PLOT NO	SUB PLOT	PALM NO	PALM QUALITY	PALM WT	FROND PROD	NO
1	DTP1	00	00	1988	1		1	0.000	788.000	22.000	
2	DTP1	00	00	1988	1		2	0.000	772.000	25.000	
3	DTP1	00	00	1988	1		3	0.000	698.000	22.000	
4	DTP1	00	00	1988	1		4	0.000	766.000	26.000	
5	DTP1	00	00	1988	1		5	0.000	746.000	21.000	
6	DTP1	00	00	1988	1		6	0.000	688.000	20.000	
7	DTP1	00	00	1988	1		7	0.000	890.000	24.000	
8	DTP1	00	00	1988	1		8	0.000	748.000	25.000	

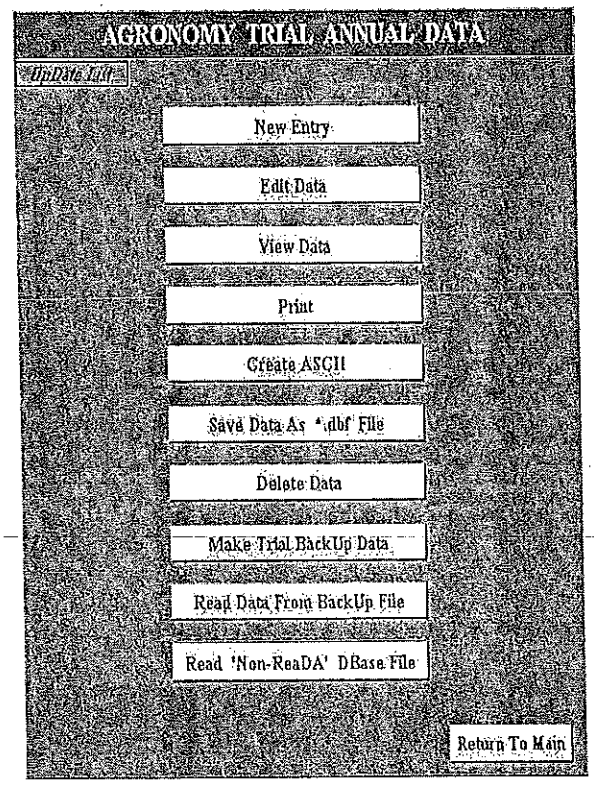
8) ***'Enabled Read Routine'*** will clear the previous data in order to read different set of data.

9) Quit the reading procedure by selecting ***'Exit (Save/Cancel)'***.

Lesson 5

{AGRONOMY TRIAL ANNUAL DATA} F.F.B YIELD, LEAF NUTRIENTS, VEGETATIVE MEASUREMENTS, SOIL NUTRIENTS.

Figure 35: Agronomy Trial Annual Data



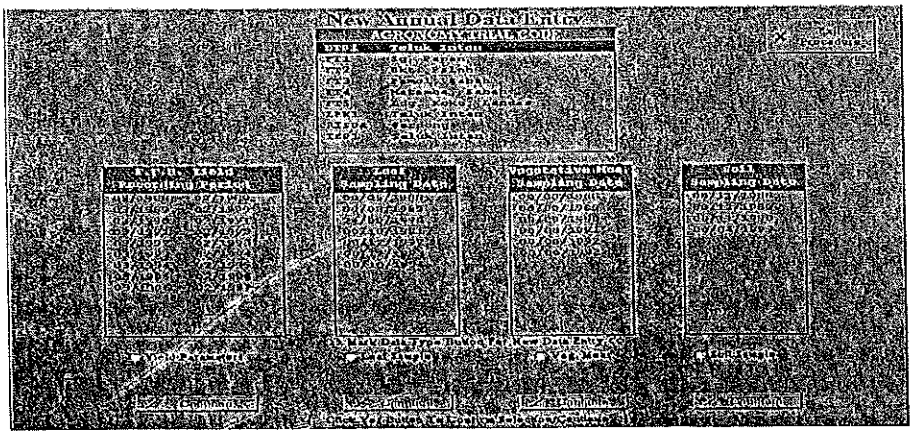
5.1 Entering New Data

Steps:

1) Click *'New Entry'* button.

- The *'New Annual Data Entry'* will appear as shown in Figure 36.

Figure 36: New Annual Data Entry



- Code selection form appear for every procedure as in Figure 36.

2) Select *Agronomy Trial Code:*

e.g. DTP1 Teluk Intan

3) Click to select the form needed out of the four types of data entry forms, as follows:

- ***Yield Parameters*** for F.F.B Yield data
- ***Leaf Sample*** for Leaf Nutrients data
- ***Veg. Meas*** for Vegetative Measurement data
- ***Soil Sample*** for Soil Nutrients data

For example, you need to enter the data for **F.F.B Yield.**

4) Mark ***'Yield Parameters'*** radio button.

- The blank box will appear for a new recording period entry.

5) Type in a new recording period.

e.g. 03/1993 - 02/1994

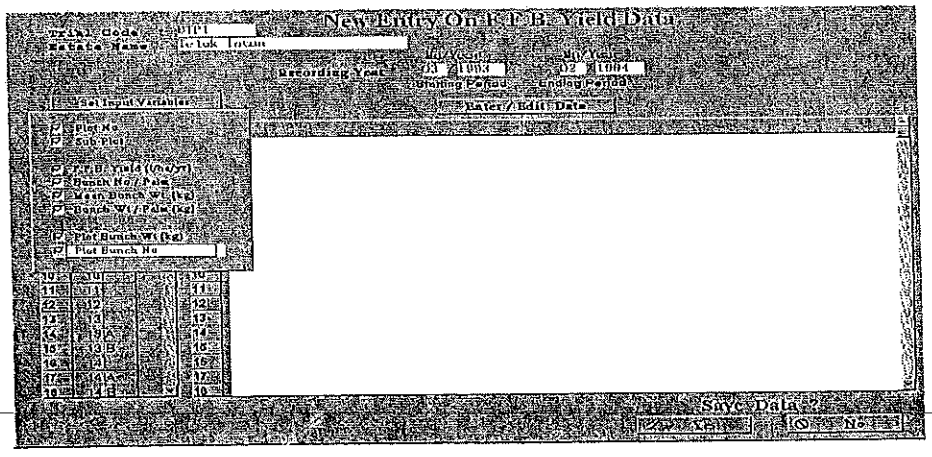
6) Click ***'Continue'*** button to proceed with the procedure.

- A new entry form will appear.

7) Click ***'Set Input Variables'*** button.

- The listed variables that appear are Plot No, Sub Plot, F.F.B Yield (t/ha/yr),Bunch No/Palm, Mean Bunch Wt(kg),Bunch Wt/Palm(kg),Plot Bunch Wt(kg) and Plot Bunch No. Mark the particular variables required and the screen will look like Figure 37.

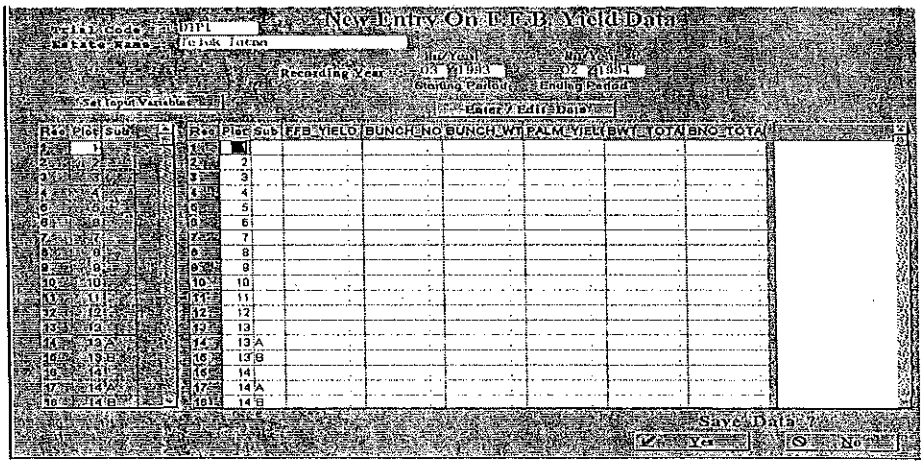
Figure 37: The empty form of New Entry On F.F.B Yield Data.



8) Click *'Enter/Edit Data'* and start your data entries.

- The form that appears is *'New Entry On F.F.B Yield Data'*. (See Figure 37.1)

Figure 37.1: New Entry On F.F.B Yield Data



9) Save your data record by clicking *'Yes'* button otherwise click *'No'* button.

If you enter different type of data, for example, leaf nutrients, vegetative measurement, and soil nutrients, the procedures are similar for each type. It only varies in marking the radio button. When you click 'Continue' button, different form title appears and when you click 'Set Input Variables', listed variables appears.

Listed below are the different form titles appearing for each type of data and its 'Set Input Variable' displaying leaf nutrients, vegetative measurement and soil nutrients respectively.

Leaf Nutrients

- Mark '*Leaf Sample*' radio button.
- Enter new sampling date e.g. 00/12/1993
- Form title appears being *'New Entry On Leaf Nutrients Data'*.
- The listed variables would be like those in Figure 38.

Vegetative Measurement

- Mark '*Veg. Meas.*' radio button.
- Enter new sampling date e.g. 00/00/1993
- Form title being *'New Entry On Vegetative Measurement'*.
- The listed variables would be like those in Figure 39.

Soil Nutrients

- Mark '*Soil Sample*' radio button.
- Enter new sampling date e.g. 00/04/1993
- Form title being *'New Entry On Soil Nutrients Data'*.
- The listed variables with additional columns (Avenue & Circle) would be like those in Figure 40.

Figure 38

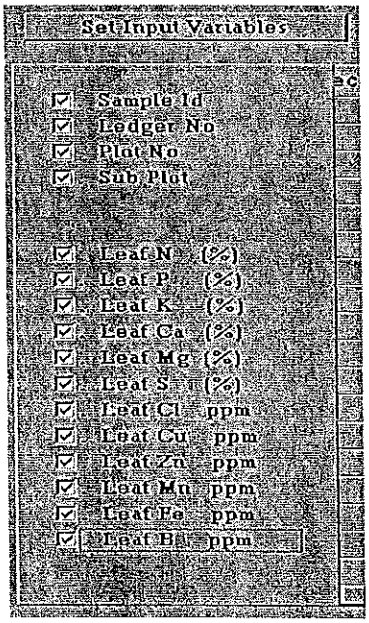


Figure 39

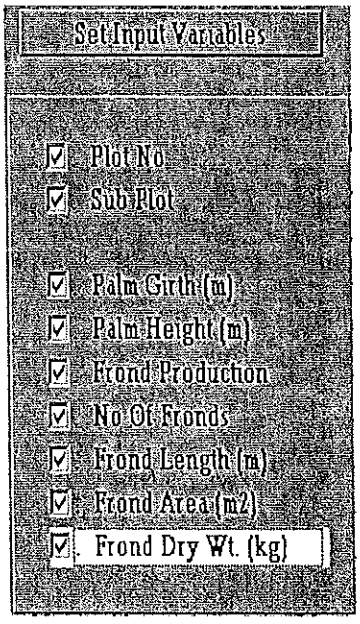
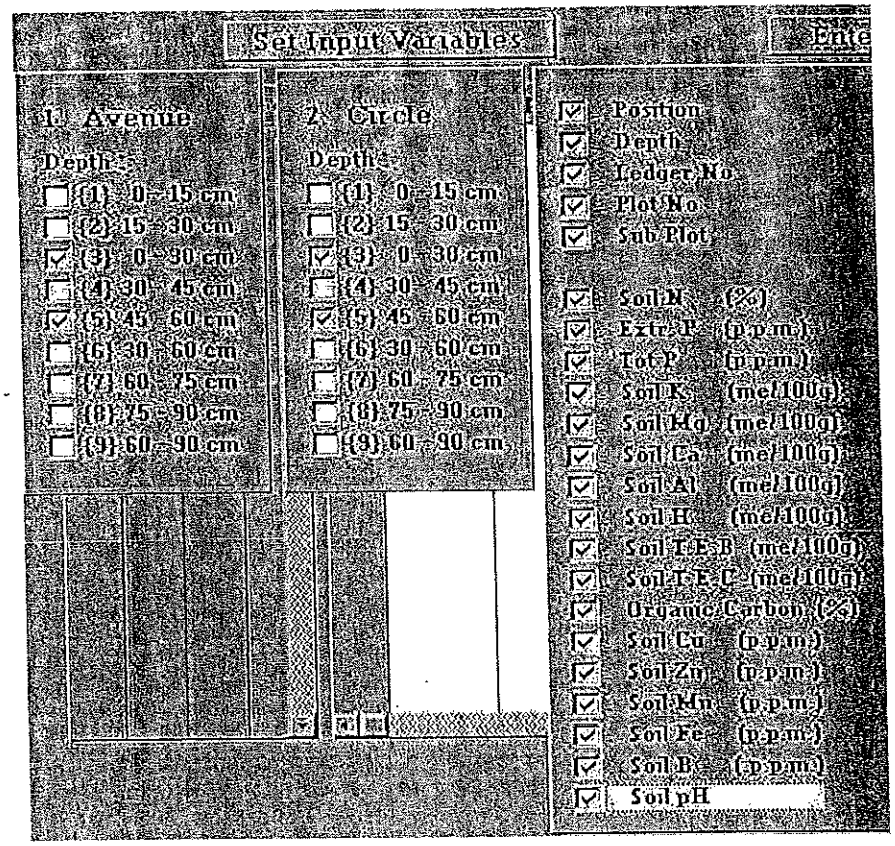


Figure 40



5.2 Editing Data Record.

Steps:

1) Click *'Edit Data'* button.

- *'Edit Annual Data'* will appear.

2) Select *Agronomy Trial Code.* e.g. DTP1 Teluk Intan

For example, you need to edit the data record for **F.F.B Yield**

3) Highlight the required period to be edited.

e.g. *F.F.B Yield Recording Period*

03/1993 - 02/1994

4) Click *'Yes'* button to continue with the procedure.

- The form that appears is *'Editing F.F.B Yield Data'.*

5) Click *'Set Input Variables'* button.

- You may vary the number of variables to be edited.

6) Click *'Enter/Edit Data'*.

- You can start editing the data record.

7) Save your data.

The procedure for editing the other data records such as leaf nutrients, vegetative measurement and soil nutrients are similar to the above procedure. However, when you click 'Yes' button, the respective form appears with the different title.

Leaf Nutrients

- Highlight *'Leaf Sampling Date'*
  e.g. 00/12/1993
- The form title is *'Editing Leaf Nutrients Data'*.

Vegetative Measurement

- Highlight *'Vegetative Mea. Sampling Date'*
  e.g. 00/00/1993
- The form title is *'Editing Vegetative Measurement Data'*.

Soil Nutrients

- Highlight *'Soil Sampling Date'*
  e.g. 00/04/1993
- The form title is *'Editing Soil Nutrients Data'*.

5.3 Viewing the Data Record

Steps:

1) Click *'View Data'* button and your screen will show the *'Display Selected Annual Data'.*

2) Select *Agronomy Trial Code*:

e.g. DTP1 Teluk Intan

For example, you want to view the data for **F.F.B Yield**

3) Click to select the period recorded.

e.g. ***F.F.B Yield Recording Period***

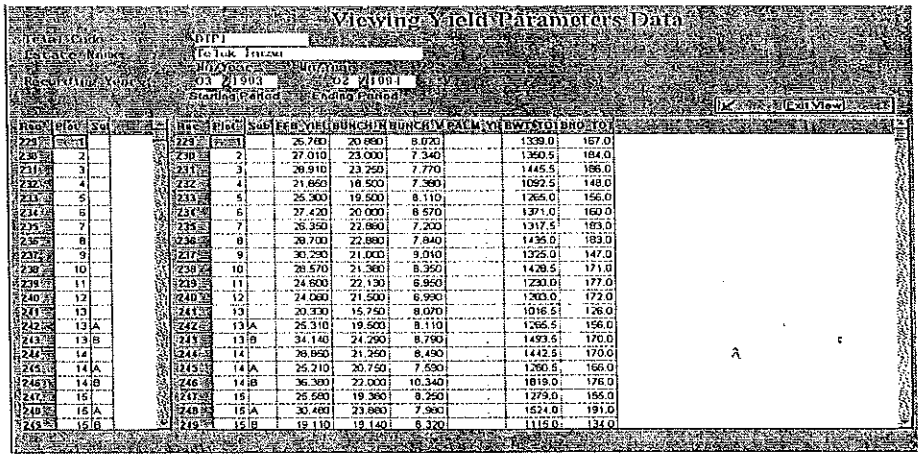
03/1993 - 02/1994

4) Click *'Yes'* button to go to the next procedure.

- *'Viewing Yield Parameters Data'* appears as in Figure 41.

5) Click *'Exit View'* button to return to main menu.

Figure 41: Viewing Yield Parameters Data

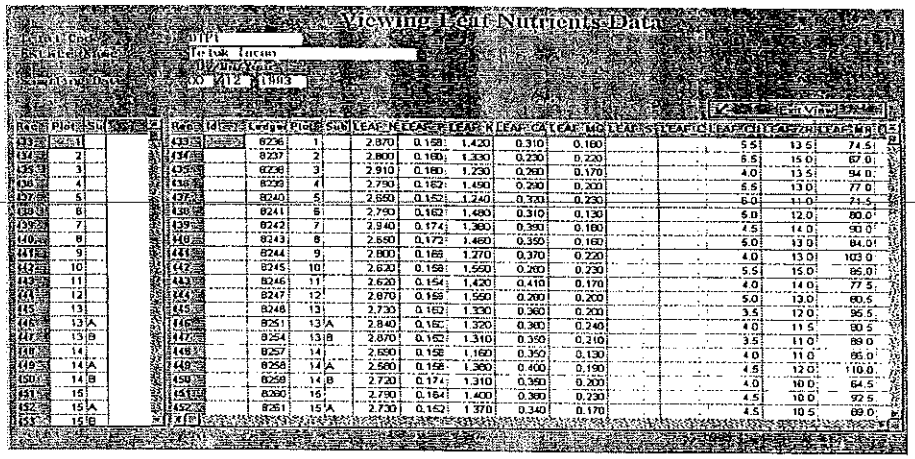


To view the other data records, the procedures will be similar.

Leaf Nutrients

- Select *'Leaf Sampling Date'* e.g. 00/12/1993
- The form title that appears will be *'Viewing Leaf Nutrients Data'* (see Figure 42).

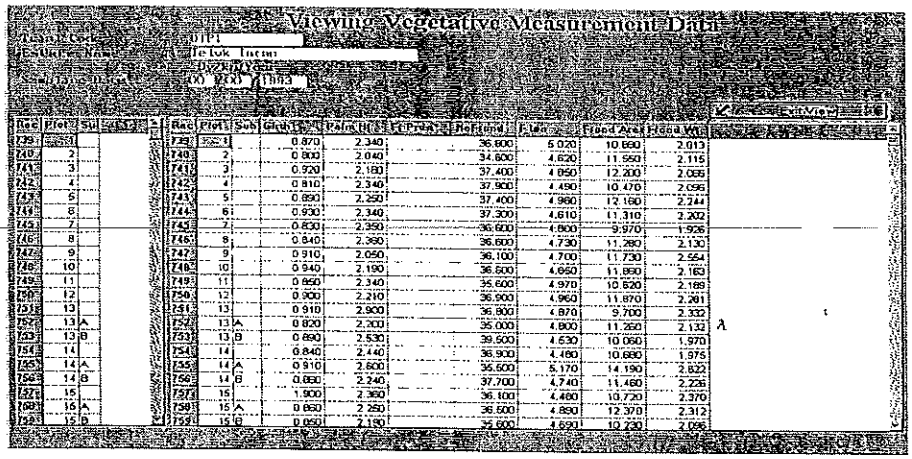
Figure 42: Viewing Leaf Nutrients Data



Vegetative Measurement

- Select *'Vegetative Mea. Sampling Date'* e.g 00/00/1993
- The form title that appears will be *'Viewing Vegetative Measurement Data'*.
  (See Figure 43)

Figure 43: Viewing Vegetative Measurement Data



Soil Nutrients

- Select *'Soil Sampling Date'* e.g. 00/04/1993
- The form title that appears will be *'Viewing Soil Nutrients Data'*
  (See Figure 44.1 and Figure 44.2).

AHMAD AFANDI B. MURDI
Pegawai Penyelidik
Stesen MPOB Lahad Datu.

Major Nutrients: The data will be used in any analysis.

Micro Elements: The data will be used in certain analysis only.

Figure 44.1: Viewing Soil Nutrients Data

(Major Nutrients)

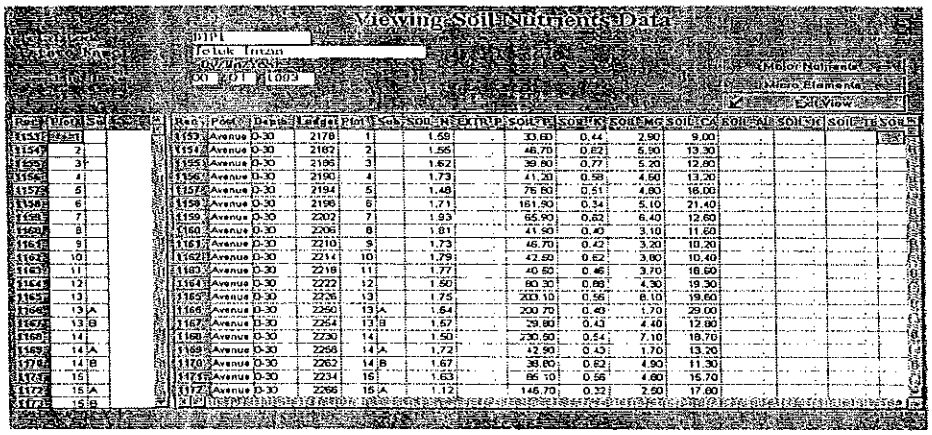
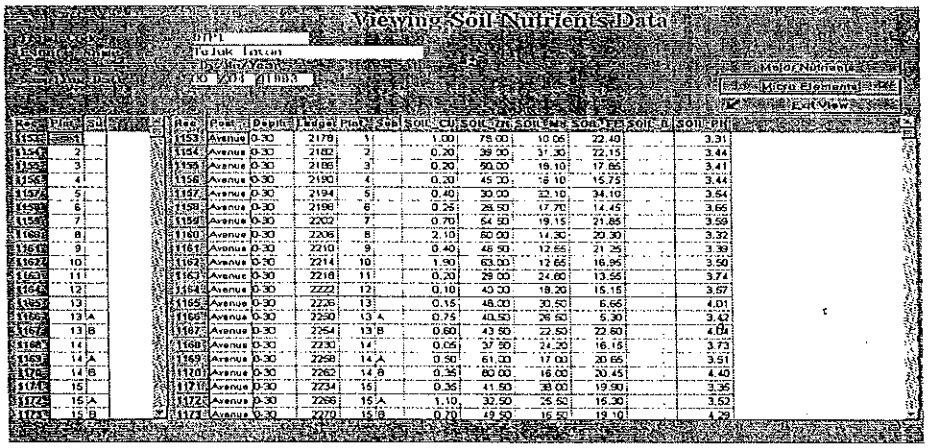


Figure 44.2: Viewing Soil Nutrients Data

(Micro Elements)



5.4 Printing the Data Record

Steps:

1) Click *'Print'* button and the screen will display *'Print Selected Annual Data'.*

2) Select *Agronomy Trial Code*

e.g. DTP1 Teluk Intan

3) Select the date required according to the type of data you want to print.

For example: **F.F.B Yield**

FFB Yield Recording Period.

e.g. 03/1993 - 02/1994

4) Click *'Yes'* button and the next form that appears will be *'Printing Yearly Yield Data'.*

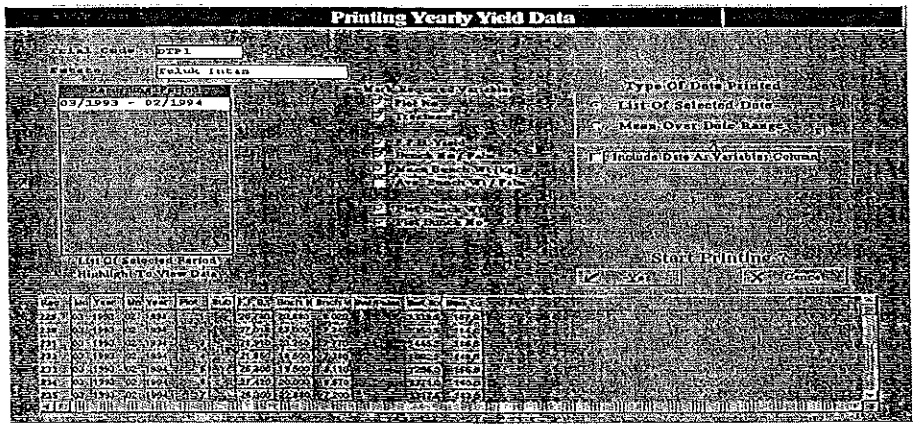
5) Highlight the required date in the *'Recording Period'* if you need to view the variables contained in the data records.

6) *'Mark Required Variables'* by clicking the selection box next to the variables.

7) *'Type of Data Printed'*: Select the printing method.

- Your screen would look like Figure 45.

Figure 45: Printing Yearly Yield Data



8) Start printing by clicking *'Yes'* button otherwise click *'Cancel'* to cancel printing.

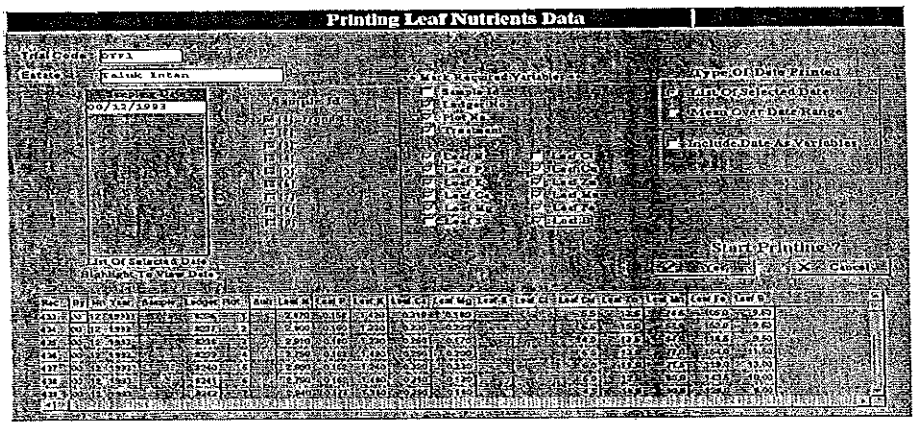
- Example of printed report in Appendix II.

To print the data records for leaf nutrients, vegetative measurement or soil nutrients, you need to select the sampling date for the data you want to print.

Leaf Nutrients

- Select *'Leaf Sampling Date'* e.g. 00/12/1993
- The form that appears will be *'Printing Leaf Nutrients Data'* (See Figure 46).
  - Example of printed report in Appendix III.

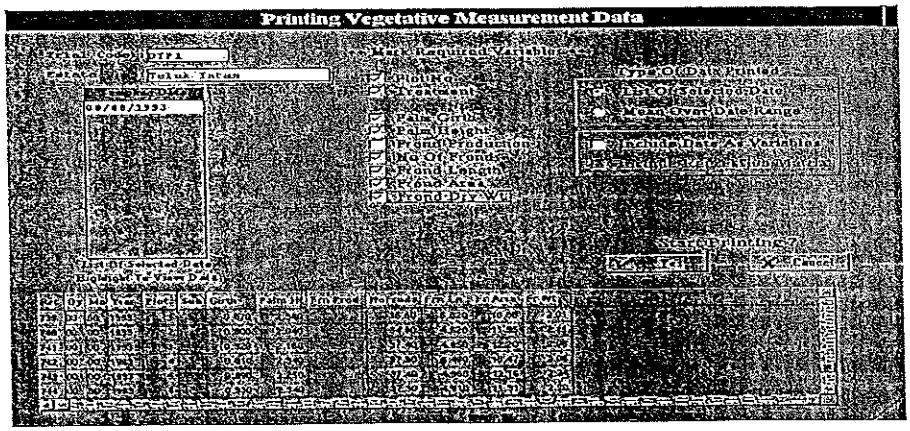
Figure 46: Printing Leaf Nutrients Data



Vegetative Measurement

- Select *'Vegetative Mea. Sampling Date'* e.g. 00/00/1993
- The form title that appears will be *'Printing Vegetative Measurement Data'* (See Figure 47).
  - Example of printed report in Appendix IV.

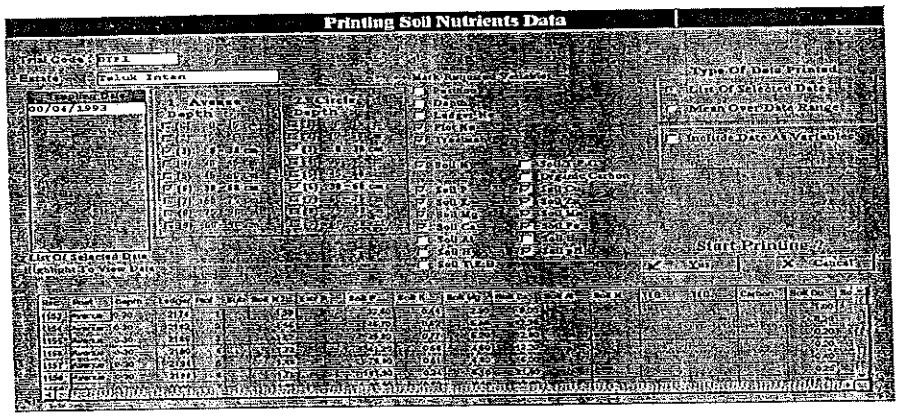
Figure 47: Printing Vegetative Measurement Data



Soil Nutrients

- Select 'Soil Sampling Date' e.g. 00/04/1993
- The form title that appears will be *'Printing Soil Nutrients Data'* (See Figure 48)
  - Example of printed report in Appendix V.

Figure 48: Printing Soil Nutrients Data



5.5 Creating an ASCII File

The procedure of creating an ASCII file for the annual trial data is essential for you in order to make the data analysis using the SAS program.

Steps:

1) Click *'Create ASCII'* button
   - The date selection frame that appears is *'Create Annual Data ASCII File'.*

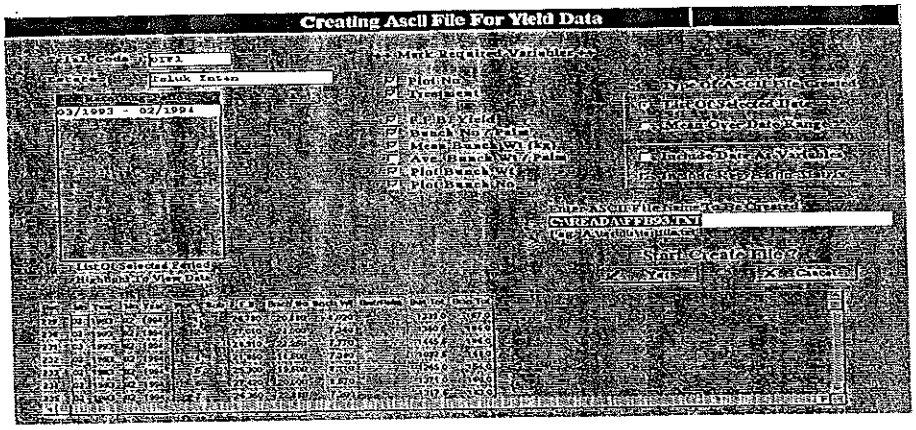
2) Select *Agronomy Trial Code*
   e.g. DTP1 Teluk Intan

3) Select the recorded period/date for the data needed.
   e.g. *F.F.B Yield*
   Recording Period 03/1993 - 02/1994

4) Click *'Yes'* button to continue with the procedure.
   - Your screen would look like Figure 49.

(*Creating ASCII File For Yield Data*)

Figure 49: Creating ASCII File For Yield Data



5) Click *'Select Recording Period'* if you need to view the data record.

6) Mark the required variables.

7) Select *'Type Of ASCII File Created'*.

8) Enter *ASCII File Name* with its drive and the directory.

e.g. C:\READA\FFB93.TXT

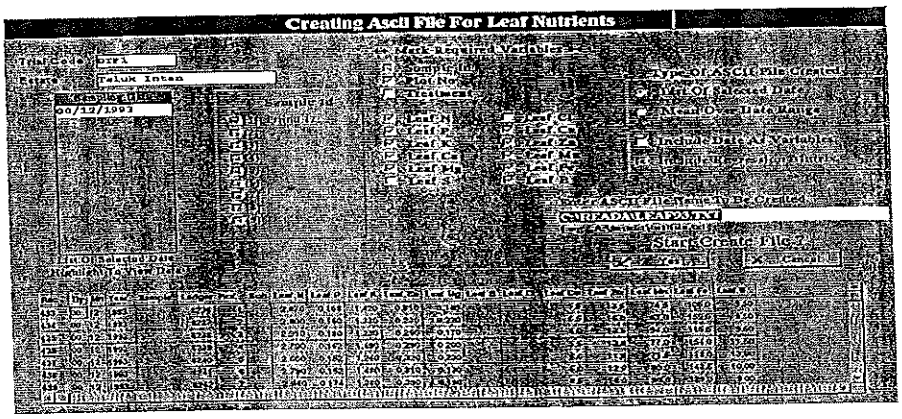
9) Confirm your creating an ASCII file by clicking *'Yes'* button, otherwise click *'Cancel'* button.

The creation of an ASCII file for other data type such as leaf nutrients, vegetative measurement and soil nutrients will be similar. The only difference is the variable contents.

Leaf Nutrients

- Select *'Leaf sampling Date'* e.g. 00/12/1993
- The form title is *'Creating ASCII File For Leaf Nutrients'* and would look like Figure 50.
- Enter *ASCII File Name* e.g. C:\READA\LEAF93.TXT

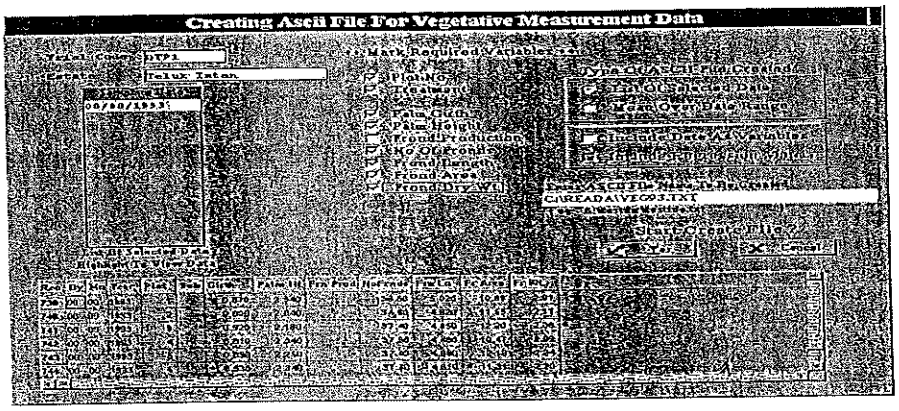
Figure 50: Creating ASCII File For Leaf Nutrients



Vegetative Measurement

- Select *'Vegetative Mea. Sampling Date'* e.g. 00/00/1993
- The form title is *'Creating ASCII File For Vegetative Measurement Data'* and would look like Figure 51.
- Enter *ASCII File Name* e.g. C:\READA\VEG93.TXT

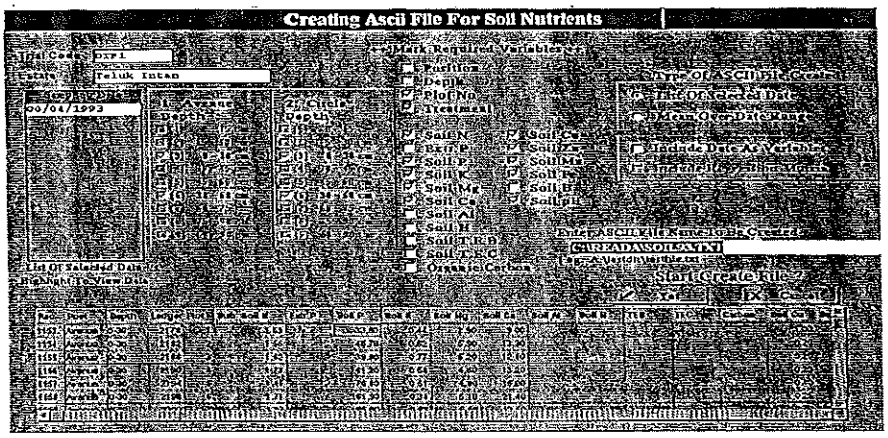
Figure 51: Creating ASCII File For Vegetative Measurement Data



Soil Nutrients

- Select *'Soil Sampling Date'* e.g. 00/04/1993
- The form title is *'Creating ASCII File For Soil Nutrients'* and would look like Figure 52.
- Enter *ASCII File Name* e.g. C:\READA\SOIL93.TXT

Figure 52: Creating ASCII File For Soil Nutrients



5.6 Saving Data As a Database File

Steps:

1) Click *'Save Data As *.dbf File'* button.

- Your screen will show *'Save Annual Data As *.dbf File'*.

2) Select *Agronomy Trial Code* needed.

3) Select desired recording period/date.

e.g. *F.F.B Yield*

FFB Yield Recording Period 03/1993 - 02/1994

4) Click *'Yes'* button to continue with the procedure.

- Your screen would look like the Figure 53.

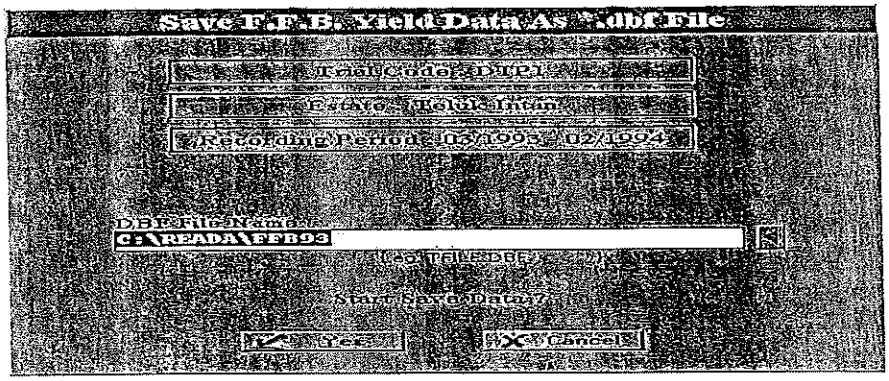
*'Save F.F.B Yield Data As *.dbf File'* will appear.

5) Enter the *DBF File Name*

e.g. C:\READA\FFB93

- Your file will automatically be saved as a *. dbf file.

Figure 53: Save F.F.B Yield Data As *.dbf File



6) Confirm your selection by clicking *'Yes'* button or otherwise, click *'Cancel'* button.

Similar steps are required in the process of saving other data types. Briefly, the information displayed in every step is as follows.

Leaf Nutrients

- Select *'Leaf Sampling Date'* e.g. 00/12/1993
- The form title is *'Save Leaf Nutrients Data As *.dbf File'.*
- Enter ***DBF File Name*** e.g. C:\READA\LEAF93.DBF (Refer to Figure 53)

Vegetative Measurement

- Select *'Vegetative Meas. Sampling Date'* e.g. 00/00/1993
- The form title is *'Save Vegetative Mea. Data As *.dbf Data'.*
- Enter ***DBF File Name*** e.g. C:\READA\VEG93.DBF (Refer to Figure 53)

Soil Nutrients

- Select *'Soil Sampling Date'* e.g. 00/04/1993
- The form title is *'Save Soil Nutrients Data As *.dbf File'.*
- Enter ***DBF File Name*** e.g. C:\READA\SOIL93.DBF (Refer to Figure 53)

5.7 Deleting Data Record

You may delete your one year data record permanently if you use this procedure.

Steps:

1) Click *'Delete Data'* button and the data selected frame as *'Delete Selected Annual Data'* will appear.

2) Select *Agronomy Trial Code*

e.g. DTP1 Teluk Intan

3) Select the recorded year to be deleted.

e.g. If you need to delete **F.F.B Yield** data record

FFB Yield Recording Period e.g. 03/1993 - 02/1994

4) Click *'Yes'* button to continue the procedure.

- Your screen will show the *'Yield Data To Be Deleted'* form.

5) Confirm your request to delete by clicking *'Yes'* button or click *'Cancel'* button to ignore it.

You can use the same procedure to delete the other data records. However, you have to ensure that the information selected before the deletion process is correct in order to avoid from making unnecessary mistakes in deleting the useful data.

5.8 Creating the Back Up File for Trial Data

This procedure is used to provide the backup file for the data records.

Steps:

1) Click *'Make Trial BackUp Data'* button.

- The date selection frame that appears will be *'Make Trial BackUp Data'*.

2) Select *Agronomy Trial Code*

e.g. DTP1 Teluk Intan

For example, you want to create a back up file for the **F.F.B Yield** data.

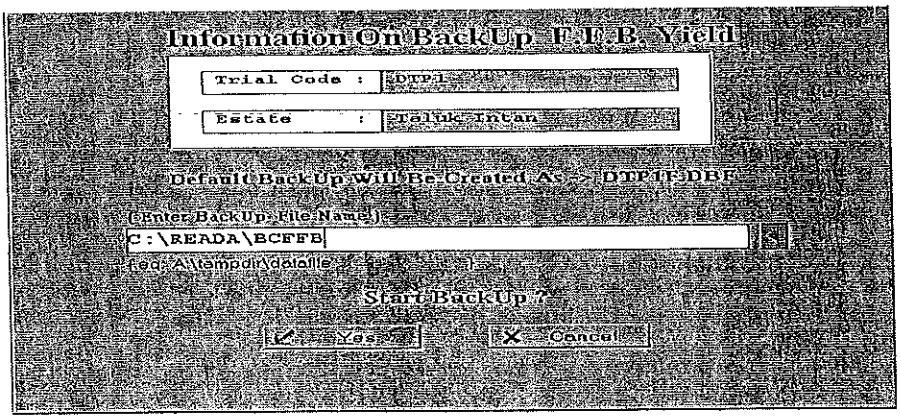
3) Mark *'BackUp Yield Data'* button and continue the procedure by clicking *'Yes'* button.

- Your screen will look like Figure 54. *'Information On Backup F.F.B Yield'.*

4) Enter your *Back Up File Name*

e.g. C:\READA\BCFFB.DBF

Figure 54: Information On BackUp F.F.B Yield



5) Click *'Yes'* button to start creating the backup file otherwise click *'Cancel'* button.

You can follow the same procedure if you want to create the back up file for the other data records. Select the desired information before you start the backup process. The following are default back up file name for each data record:

Leaf Nutrients

- Mark *'Backup Leaf Data'* radio button.
- The form title is *'Information On Back Up Leaf Data'*.
- Enter *Back Up File Name* e.g. C:\READA\BCLEAF.DBF (Refer to Figure 54)

Vegetative Measurement

- Mark *'Backup Vegetative'* radio button.
- The form title is *'Information On Back Up Vegetative Data'*.
- Enter *Back Up File Name* e.g. C:\READA\BCVEG.DBF

(Refer to Figure 54)

Soil Nutrients

- Mark *'Backup Soil Data'* radio button.
- The form title is *'Information On Back up Soil Data'*.
- Enter *Back Up File Name* e.g. C:\READA\BCSOIL.DBF

(Refer to Figure 54)

5.9 Reading Data From Back Up File

This procedure is used when you need to recover the data record for the annual trial data that you have deleted before.

Steps:

1) Click *'Read Data From Back Up File'* button.

2) Select *Agronomy Trial Code:*

e.g. DTP1 Teluk Intan

For example, you want to read the backup file for the **F.F.B Yield.**

3) Mark *'Backup Yield Data'.*

4) Click *'Yes'* button and the form that appears is *'Reading Yield Data From Backup File'.*

5) Enter your *Backup File Name* or you can browse in the directory and double-click to select the file.

e.g. C:\READA\BCFFB.DBF

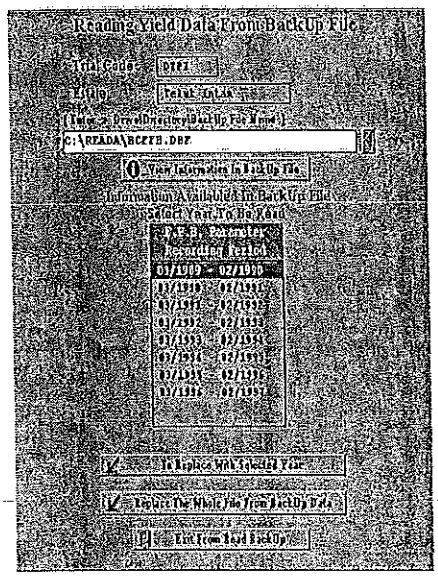
6) Click *'View Information In Backup File'* and it will show the listed period recorded.

- Now your screen would look like Figure 56.

7) *'Select Year To Be Read'* from the list.

8) Select either *'To Replace With Selected Year'* or *'Replace The Whole File From Backup File'* and your data will automatically be read and saved as a file in ReaDA file system, otherwise click *'Exit From Read Backup'* button.

Figure 56: Reading Yield Data From Backup File



Similar methods will be required to read the backup data for the leaf nutrients, vegetative measurement or soil nutrients.

Leaf Nutrients

- Click *'Backup Leaf Data'* radio button.
- The form that appears is *'Reading Leaf Nutrients Data From BackUp File'*.
- Enter *Backup File Name* e.g. C:\READA\BCLEAF.DBF

Vegetative Measurement

- Mark *'Backup Vegetative'* radio button.

- The form that appears is *'Reading Vegetative Measurement Data From BackUp File'.*
- Enter *Backup File Name* e.g. C:\READA\BCVEG.DBF

Soil Nutrients

- Mark '*Backup Soil Data*' radio button.
- The form that appears is *'Reading Soil Nutrients Data From BackUp File'.*
- Enter *Backup File Name* e.g. C:\READA\BCSOIL.DBF

5.10 Reading Record From External Database File

This procedure is used when you need to read the data that was saved as a database file by using the 'non-ReaDA' system (external database file).

Steps:

1) Click *'Read 'Non-ReaDA' DBase File'* button. The data selection form displays *'Read Annual Data From 'Non-ReaDA' Dbase File'.*

2) Select *Agronomy Trial Code*

e.g. DTP1 Teluk Intan

For example, you need to read the **F.F.B Yield** from the external database file.

3) Mark *'Yield Parameters'* radio button, then enter the *'Recording Period'* needed. e.g. 03/1994 - 02/1995

4) Continue the procedure by clicking *'Yes'* button.

- Your screen would look like Figure 57.

5) Enter the *'Dbase File Name'* or browse the directories. This is an external database file name. e.g. DTPY194.DBF

6) *First Record To Be Read:* Enter your first record you want to read.

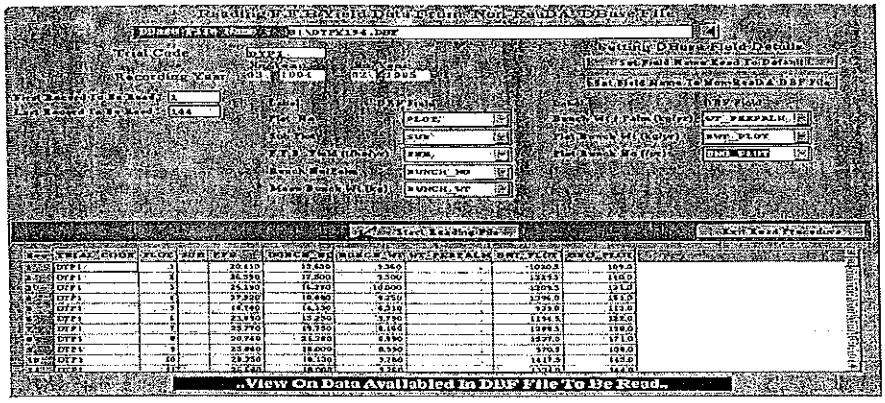
7) *Last Record To Be Read:* Enter your last record you want to read.

8) Click *'Setting Dbase Field Details'* button.

- Select the field name by clicking either *'Set Field Name Read To*

Default' button or *'Set Field Name Read To'Non-ReaDA' DBF' File'* if you want to change it according to the external database field name.

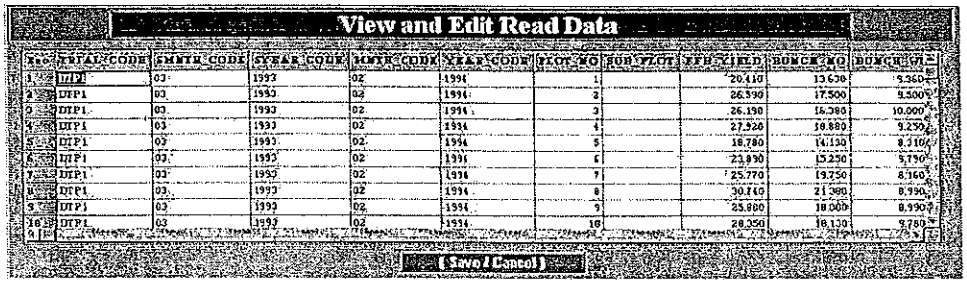
Figure 57: Reading F.F.B Yield Data From 'Non-ReaDA' DBase File



9) Click *'Start Reading File'* button otherwise click *'Exit Read Procedure'* button.

- After the 'Start Reading File' process, the browse field title will change to *'View and Edit Read Data'* as shown in Figure 57.1.

Figure 57.1 : View and Edit Read Data (F.F.B Yield)



10) Save your data record.

Similar procedure will be used for the other data records such as leaf nutrients, vegetative measurement or soil nutrients. Different information is needed for each type of data record, different form title with the different field name to select. The examples are given below:

Leaf Nutrients.

- Mark *'Leaf Sample'* radio button.
- Enter '*Leaf Sampling Date'* e.g 00/10/1994
- The form title is *'Reading Leaf Data From 'Non-ReaDA' DBase File*' as in Figure 58.
- Enter *DBase File Name*: e.g. B:\DTPL1195.DB
- *'View and Edit Read Data'* as shown in Figure 58.1.

Figure 58: Reading Leaf Data From 'Non-ReaDA' DBase File

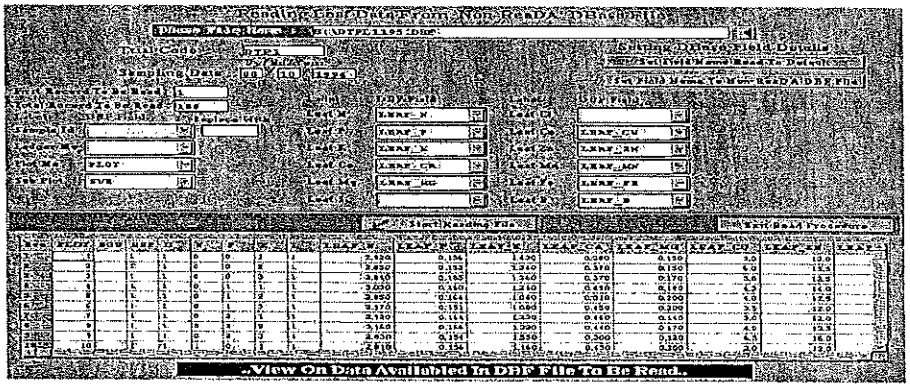
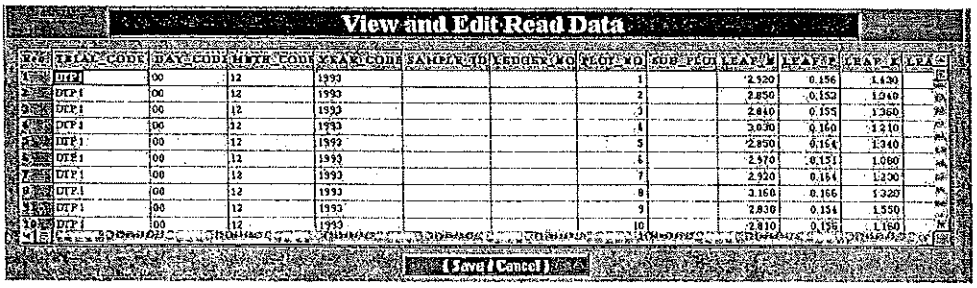


Figure 58.1: View and Edit Read Data (Leaf Nutrients)



Vegetative Measurement

- Mark *'Veg. Meas'* radio button.
- Enter *'Vegetative Mea. Sampling Date'* e.g. 00/00/1994
- The form title is *'Reading Vegetative Measurement Data From 'Non-ReaDA' DBase File'*. (see Figure 59).
- Enter *DBase File Name*: e.g. B:\DTPVM.DBF
- *'View and Edit Read Data'* as shown in Figure 59.1.

Figure 59: Reading Vegetative Measurement Data From 'Non-ReaDA' Dbase File

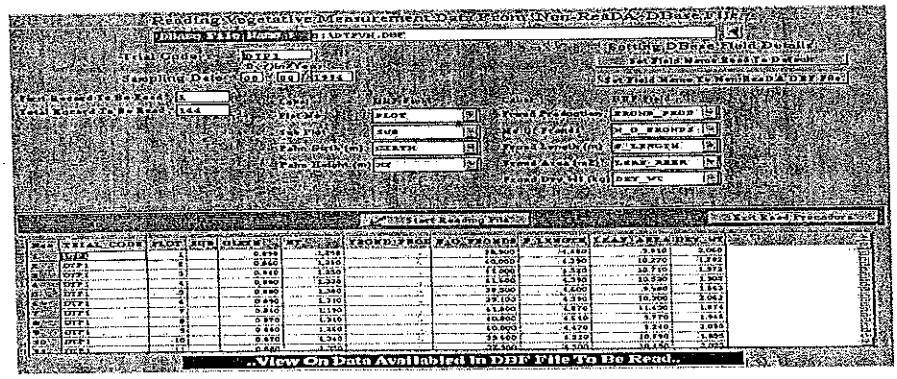
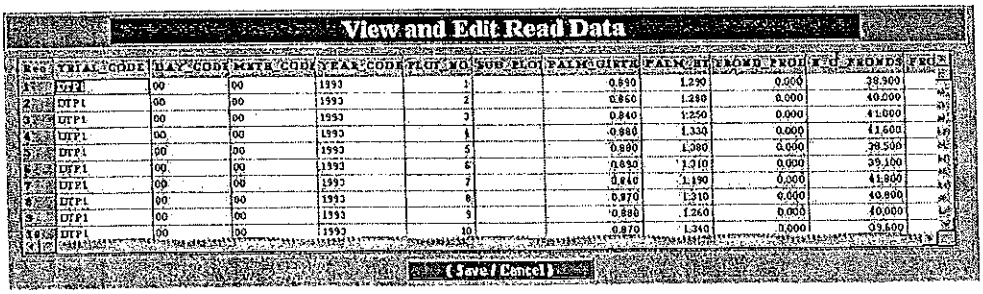


Figure 59.1: View and Edit Read Data (Vegetative Measurement)



Soil Nutrients

- Mark *'Soil Sample'* radio button.
- Enter *'Soil Sampling Date'* e.g. 00/00/1994
- The form title is *'Reading Soil Data From 'Non-ReaDA' DBase File'*. (see Figure 60).
- Enter *DBase File Name*: e.g. B:\DTPSOIL.DBF
- *'View and Edit Read Data'* as shown in figure 60.1.

Figure 60: Reading Soil Data From 'Non-ReaDA' DBase File

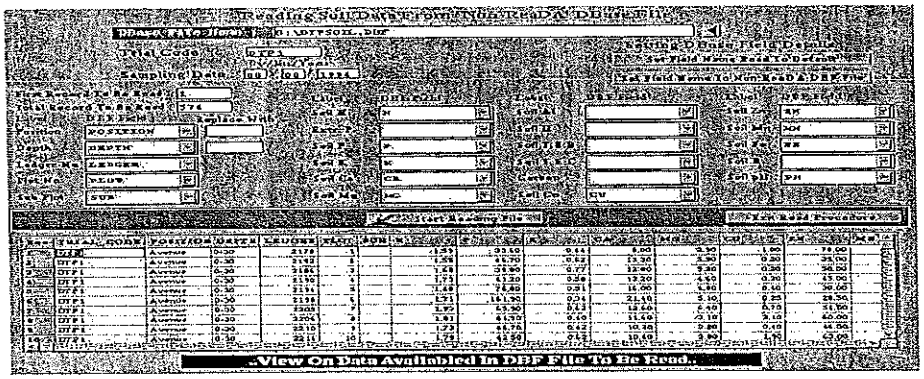
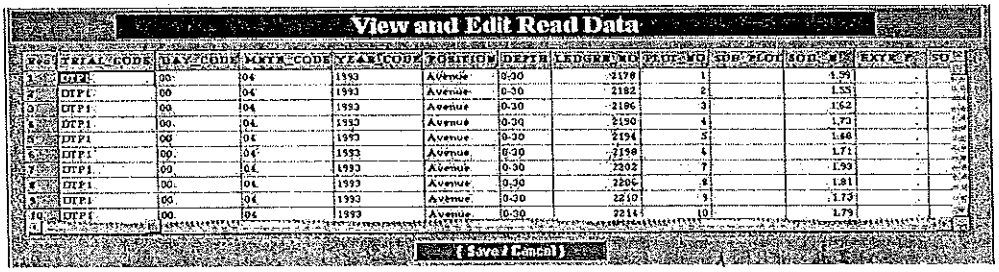


Figure 60.1: View and Edit Read Data (Soil Nutrients)



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